SRI CHANDRASEKHARENDRA SARASWATHI VISWA MAHAVIDYALAYA

(UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956)
ENATHUR, KANCHIPURAM – 631 561

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Class : III B.E. (CSE) S1 Section

Course Code: BCSF186P80

Course Name: OPEN CV LAB

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of work done by Mr/Ms._____A.Adarsh_____ , with Reg. No _____11239A003_____of III-B.E.(CSE-S1) during the academic year 2025 – 2026.

Station:
Date:

Staff-in-charge **Head of the Department**

Submitted for the Practical examination held on ____30-04-2026_____.

Examiner-1 **Examiner-2**

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Ex. No.: 01 **READ AND DISPLAY IMAGE**

Date: 27.01.2026

AIM:

To read an image from disk and display it in a window using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Use cv2.imread('image.jpg') to load the image from the disk into memory.
3. Check that the image was loaded successfully (it should not be None).
4. Use cv2.imshow('Image', img) to open a display window and show the image.
5. Call cv2.waitKey(0) to pause execution and keep the window open until a key is pressed.
6. Call cv2.destroyAllWindows() to close all open display windows.

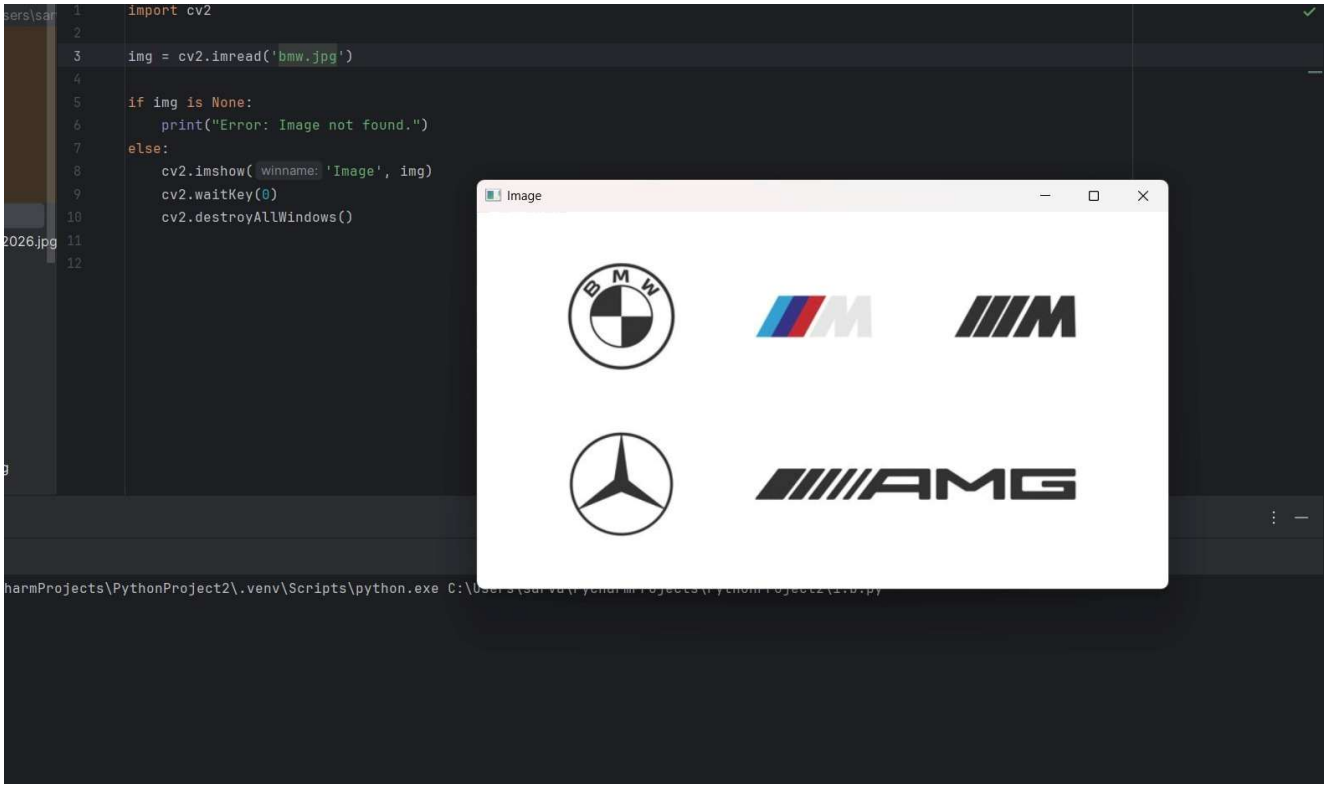
CODE:

```
import cv2

img = cv2.imread('image.jpg')

if img is None:
    print("Error: Image not found.")
else:
    cv2.imshow('Image', img)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

RESULT:

The image was successfully read from disk and displayed in an OpenCV window.

Output :


import cv2
img = cv2.imread('bmw.jpg')
if img is None:
print("Error: Image not found.")
else:
cv2.imshow(winname: 'Image', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
Image
AMG

Ex. No.: 02 **DISPLAY PROPERTIES OF IMAGE**

Date: 27.01.2026

AIM:

To read an image and display its properties such as shape, size, and data type.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image using cv2.imread('image.jpg').
3. Print img.shape to get the image dimensions (height, width, number of channels).
4. Print img.size to get the total number of pixel elements.
5. Print img.dtype to get the data type of the pixel values.
6. Observe the output in the console.

CODE:

```
import cv2

img = cv2.imread('image.jpg')

print("Shape (Height x Width x Channels):", img.shape)
print("Total Size (pixels):", img.size)
print("Data Type:", img.dtype)
```

RESULT:

The shape, size, and data type of the image were successfully retrieved and displayed.

Output :

```python
import cv2

img = cv2.imread('bmw.jpg')

print("Shape (Height x Width x Channels):", img.shape)
print("Total Size (pixels):", img.size)
print("Data Type:", img.dtype)
```

Run 1.b

```
C:\Users\sarva\PycharmProjects\PythonProject2\.venv\Scripts\python.exe C:\Users\sarva\PycharmProjects\PythonProject2\1.b.py
Shape (Height x Width x Channels): (360, 661, 3)
Total Size (pixels): 713880
Data Type: uint8

Process finished with exit code 0
```

Ex. No.: 03 **CONVERT IMAGE TO GRAYSCALE**

Date: 27.01.2026

AIM:

To convert a color (BGR) image to a grayscale image using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the color image using cv2.imread('image.jpg').
3. Use cv2.cvtColor(img, cv2.COLOR_BGR2GRAY) to convert the image to grayscale.
4. Display the grayscale image using cv2.imshow('Grayscale', gray).
5. Call cv2.waitKey(0) to keep the window open.
6. Close all windows using cv2.destroyAllWindows().

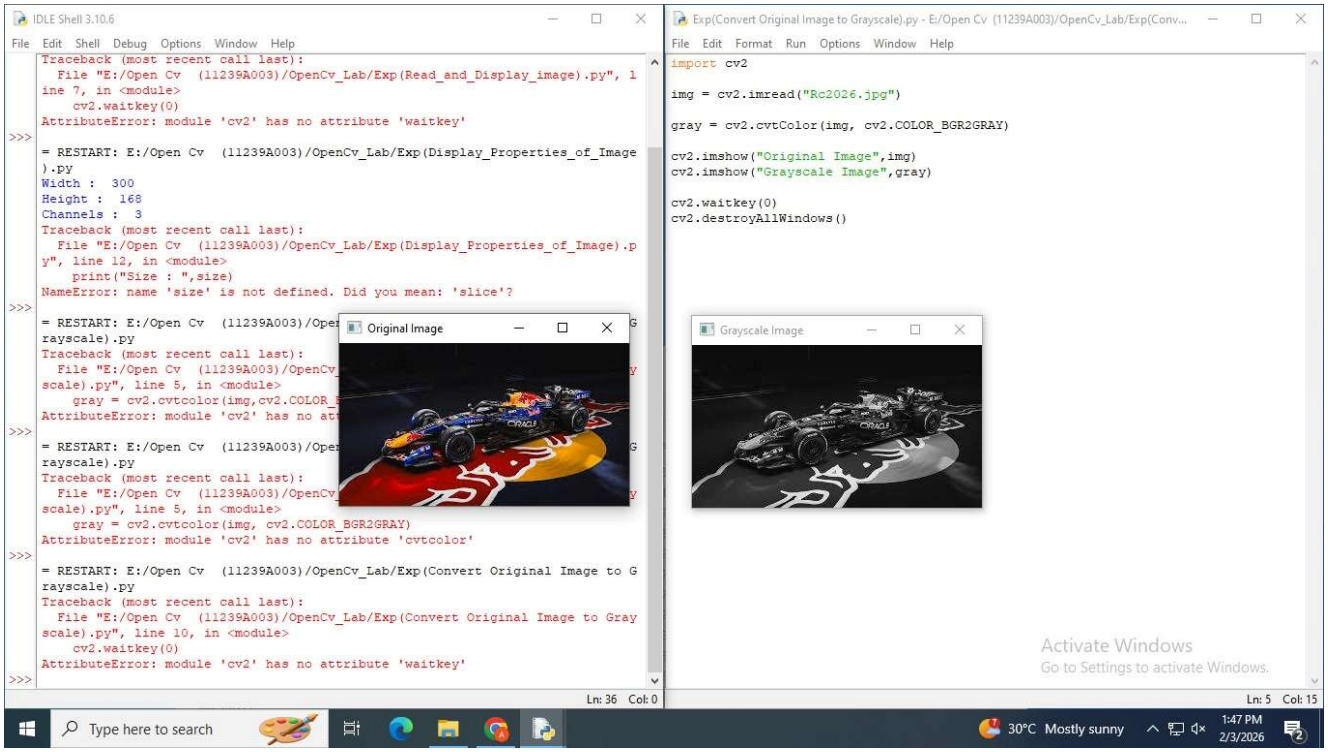
CODE:

```
import cv2

img = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

cv2.imshow('Grayscale Image', gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The color image was successfully converted to grayscale and displayed.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Read_and_Display_image).py", line 7, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Display_Properties_of_Image).py
Width : 300
Height : 168
Channels : 3
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Display_Properties_of_Image).py", line 12, in <module>
print("Size : ",size)
NameError: name 'size' is not defined. Did you mean: 'slice'?
>>>
= RESTART: E:/Open Cv (11239A003)/Ope
rayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv
scale).py", line 5, in <module>
gray = cv2.cvtcolor(img,cv2.COLOR_
AttributeError: module 'cv2' has no att
>>>
= RESTART: E:/Open Cv (11239A003)/Ope
rayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv
scale).py", line 5, in <module>
gray = cv2.cvtcolor(img, cv2.COLOR_BGR2GRAY)
AttributeError: module 'cv2' has no attribute 'cvtcolor'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py", line 10, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
Ln: 36 Col: 0
Original Image
Exp(Convert Original Image to Grayscale).py - E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Conv...
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
cv2.imshow("Original Image",img)
cv2.imshow("Grayscale Image",gray)
cv2.waitkey(0)
cv2.destroyAllWindows()
Grayscale Image
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 15
Type here to search
30°C Mostly sunny
1:47 PM
2/3/2026

Ex. No.: 04 **CONVERT IMAGE TO HSV**

Date: 02.02.2026

AIM:

To convert a color image from BGR to HSV (Hue, Saturation, Value) color space using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the color image using cv2.imread('image.jpg').
3. Use cv2.cvtColor(img, cv2.COLOR_BGR2HSV) to convert the image to HSV color space.
4. Display the HSV image using cv2.imshow('HSV Image', hsv).
5. Call cv2.waitKey(0) to keep the window open.
6. Close all windows using cv2.destroyAllWindows().

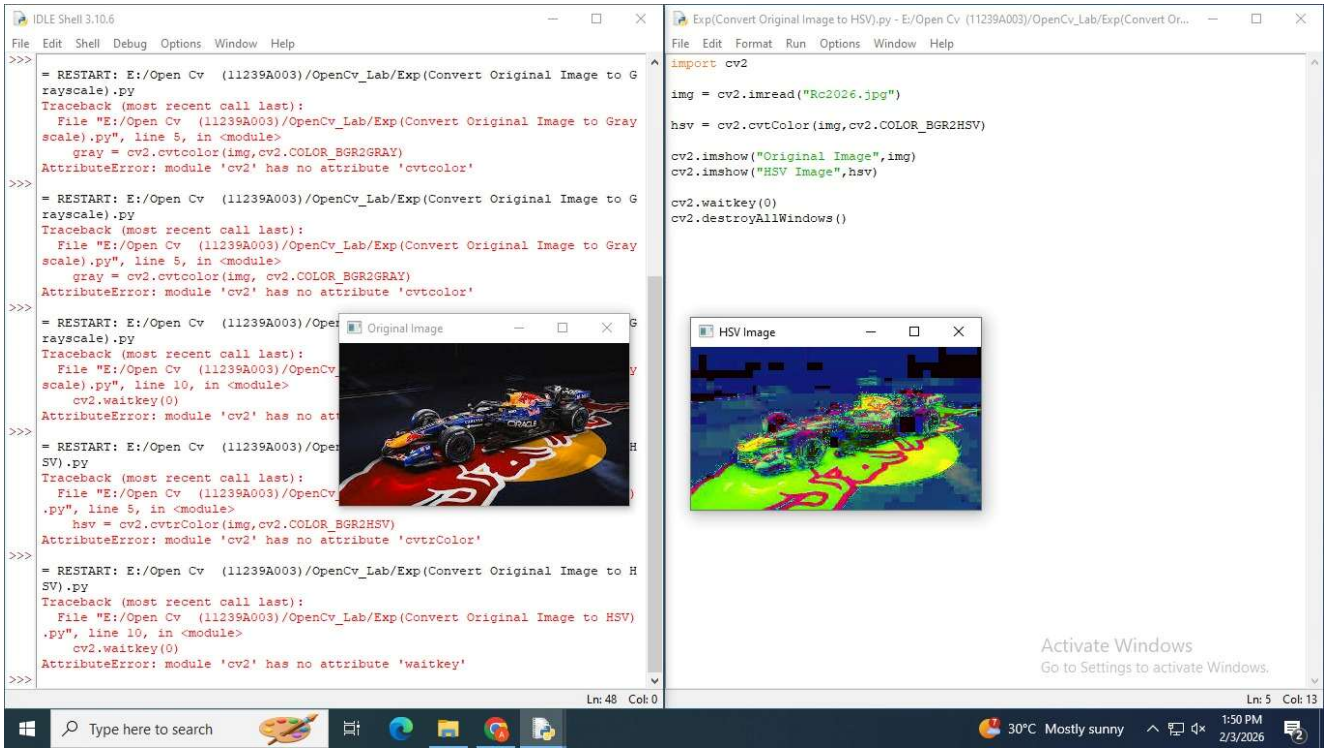
CODE:

```
import cv2

img = cv2.imread('image.jpg')
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

cv2.imshow('HSV Image', hsv)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The image was successfully converted from BGR to HSV color space and displayed.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py", line 5, in <module>
gray = cv2.cvtcolor(img,cv2.COLOR_BGR2GRAY)
AttributeError: module 'cv2' has no attribute 'cvtcolor'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to Grayscale).py", line 5, in <module>
gray = cv2.cvtcolor(img, cv2.COLOR_BGR2GRAY)
AttributeError: module 'cv2' has no attribute 'cvtcolor'
>>>
= RESTART: E:/Open Cv (11239A003)/Open
rayscale).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv
scale).py", line 10, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no att
>>>
= RESTART: E:/Open Cv (11239A003)/Open
SV).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv
.py", line 5, in <module>
hsv = cv2.cvtrColor(img,cv2.COLOR_BGR2HSV)
AttributeError: module 'cv2' has no attribute 'cvtrColor'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py", line 10, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
Ln: 48 Col: 0
Original Image
Exp(Convert Original Image to HSV).py - E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Or...
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
hsv = cv2.cvtColor(img,cv2.COLOR_BGR2HSV)
cv2.imshow("Original Image",img)
cv2.imshow("HSV Image",hsv)
cv2.waitkey(0)
cv2.destroyAllWindows()
HSV Image
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 13
Type here to search
30°C Mostly sunny
1:50 PM
2/3/2026

Ex. No.: 05 **ADD TEXT ON IMAGE**

Date: 02.02.2026

AIM:

To overlay text on an image using OpenCV's cv2.putText() function.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image using cv2.imread('image.jpg').
3. Select a font style using cv2.FONT_HERSHEY_SIMPLEX.
4. Use cv2.putText() with parameters: image, text string, position (x, y), font, font scale, color (B, G, R), and thickness.
5. Display the result using cv2.imshow().
6. Wait for a key press and close all windows.

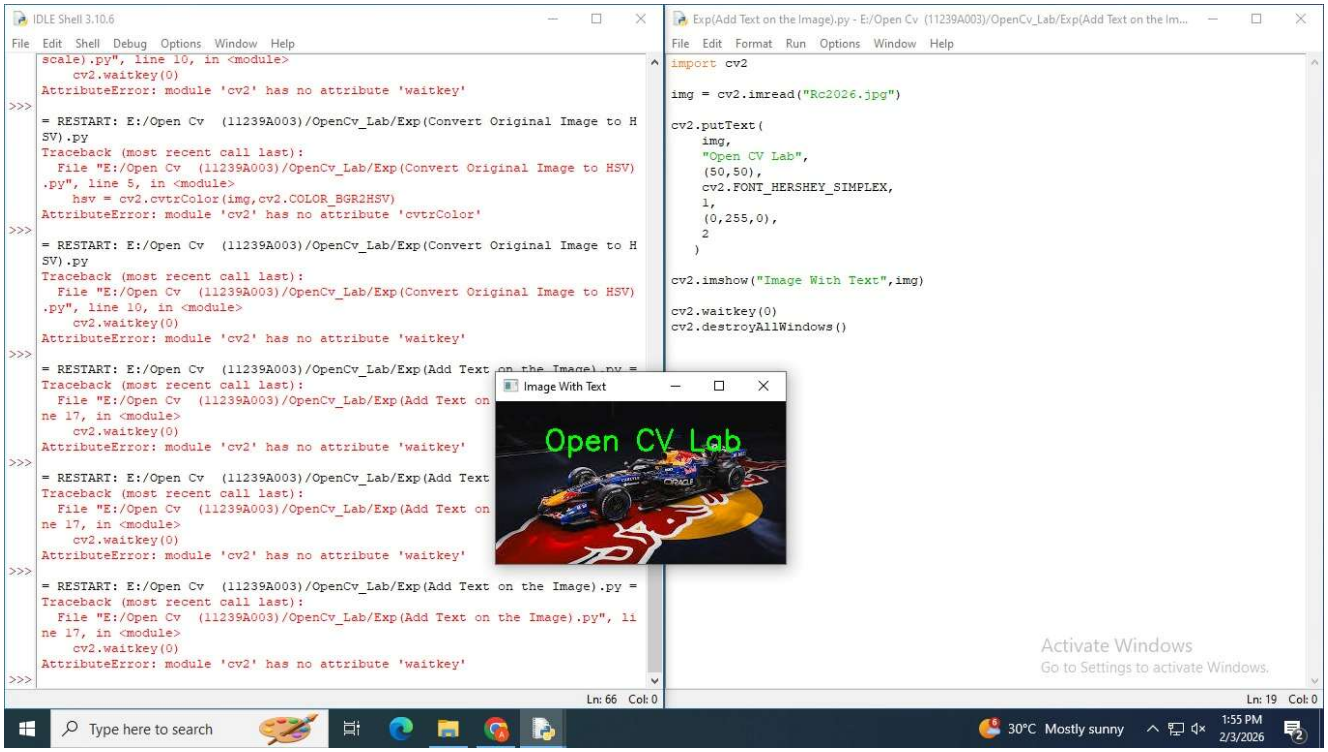
CODE:

```
import cv2

img = cv2.imread('image.jpg')
font = cv2.FONT_HERSHEY_SIMPLEX

cv2.putText(img, 'OpenCV Lab', (50, 50), font, 1.2, (0, 255, 0), 2)

cv2.imshow('Image with Text', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Text was successfully overlaid on the image at the specified position using cv2.putText().

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
scale).py", line 10, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py", line 5, in <module>
hsv = cv2.cvtrColor(img,cv2.COLOR_BGR2HSV)
AttributeError: module 'cv2' has no attribute 'cvtrColor'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Convert Original Image to HSV).py", line 10, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Add Text on the Image).py =
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Add Text on the Image).py", line 17, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
= RESTART: E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Add Text on the Image).py =
Traceback (most recent call last):
File "E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Add Text on the Image).py", line 17, in <module>
cv2.waitkey(0)
AttributeError: module 'cv2' has no attribute 'waitkey'
>>>
Ln: 66 Col: 0
Exp(Add Text on the Image).py - E:/Open Cv (11239A003)/OpenCv_Lab/Exp(Add Text on the Im...
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
cv2.putText(
img,
"Open CV Lab",
(50,50),
cv2.FONT_HERSHEY_SIMPLEX,
1,
(0,255,0),
2
)
cv2.imshow("Image With Text",img)
cv2.waitkey(0)
cv2.destroyAllWindows()
Image With Text
Open CV Lab
Activate Windows
Go to Settings to activate Windows.
Ln: 19 Col: 0
Type here to search
30°C Mostly sunny
1:55 PM
2/3/2026

Ex. No.: 06 **DRAW DIFFERENT SHAPES**

Date: 02.02.2026

AIM:

To draw basic shapes such as lines, circles, rectangles, and ellipses on a blank image using OpenCV.

PROCEDURE:

1. Import cv2 and numpy.
2. Create a blank black image using np.zeros((400, 400, 3), dtype=np.uint8).
3. Draw a line using cv2.line() by specifying start and end points.
4. Draw a circle using cv2.circle() by specifying center, radius, and color.
5. Draw a rectangle using cv2.rectangle() with top-left and bottom-right corners.
6. Draw an ellipse using cv2.ellipse() and display the final image.

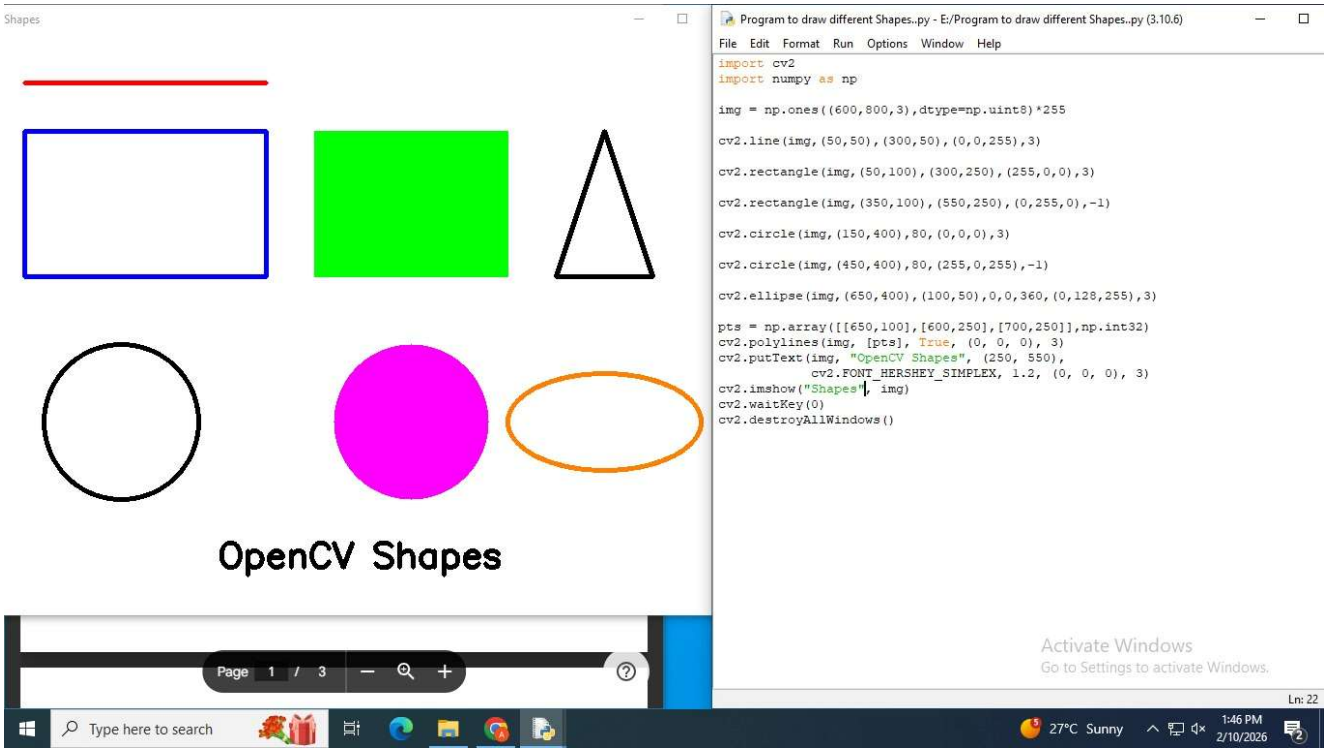
CODE:

```
import cv2
import numpy as np

img = np.zeros((400, 400, 3), dtype=np.uint8)

cv2.line(img, (0, 0), (400, 400), (0, 255, 0), 2)
cv2.circle(img, (200, 200), 80, (0, 0, 255), 3)
cv2.rectangle(img, (50, 50), (180, 150), (255, 0, 0), 2)
cv2.ellipse(img, (300, 100), (60, 30), 0, 0, 360, (255, 255, 0), 2)

cv2.imshow('Shapes', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

A line, circle, rectangle, and ellipse were successfully drawn on a blank image.

Output :


Shapes
OpenCV Shapes
Program to draw different Shapes..py - E:/Program to draw different Shapes..py (3.10.6)
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = np.ones((600,800,3),dtype=np.uint8)*255
cv2.line(img,(50,50),(300,50),(0,0,255),3)
cv2.rectangle(img,(50,100),(300,250),(255,0,0),3)
cv2.rectangle(img,(350,100),(550,250),(0,255,0),-1)
cv2.circle(img,(150,400),80,(0,0,0),3)
cv2.circle(img,(450,400),80,(255,0,255),-1)
cv2.ellipse(img,(650,400),(100,50),0,0,360,(0,128,255),3)
pts = np.array([[650,100],[600,250],[700,250]],np.int32)
cv2.polylines(img, [pts], True, (0, 0, 0), 3)
cv2.putText(img, "OpenCV Shapes", (250, 550),
cv2.FONT_HERSHEY_SIMPLEX, 1.2, (0, 0, 0), 3)
cv2.imshow("Shapes", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 22
Page 1 / 3
Type here to search
27°C Sunny
1:46 PM
2/10/2026

Ex. No.: 07 **READ AND DISPLAY VARIOUS IMAGE FORMATS**

Date: 09.02.2026

AIM:

To read and display image files in different formats (JPG, PNG, BMP, TIFF) using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Create a list of image filenames with different extensions: .jpg, .png, .bmp, .tiff.
3. Use a loop to iterate through the list.
4. For each file, use cv2.imread() to load the image.
5. Check if the image was loaded successfully before displaying.
6. Display each image using cv2.imshow() and wait for a key press; close all windows after the loop.

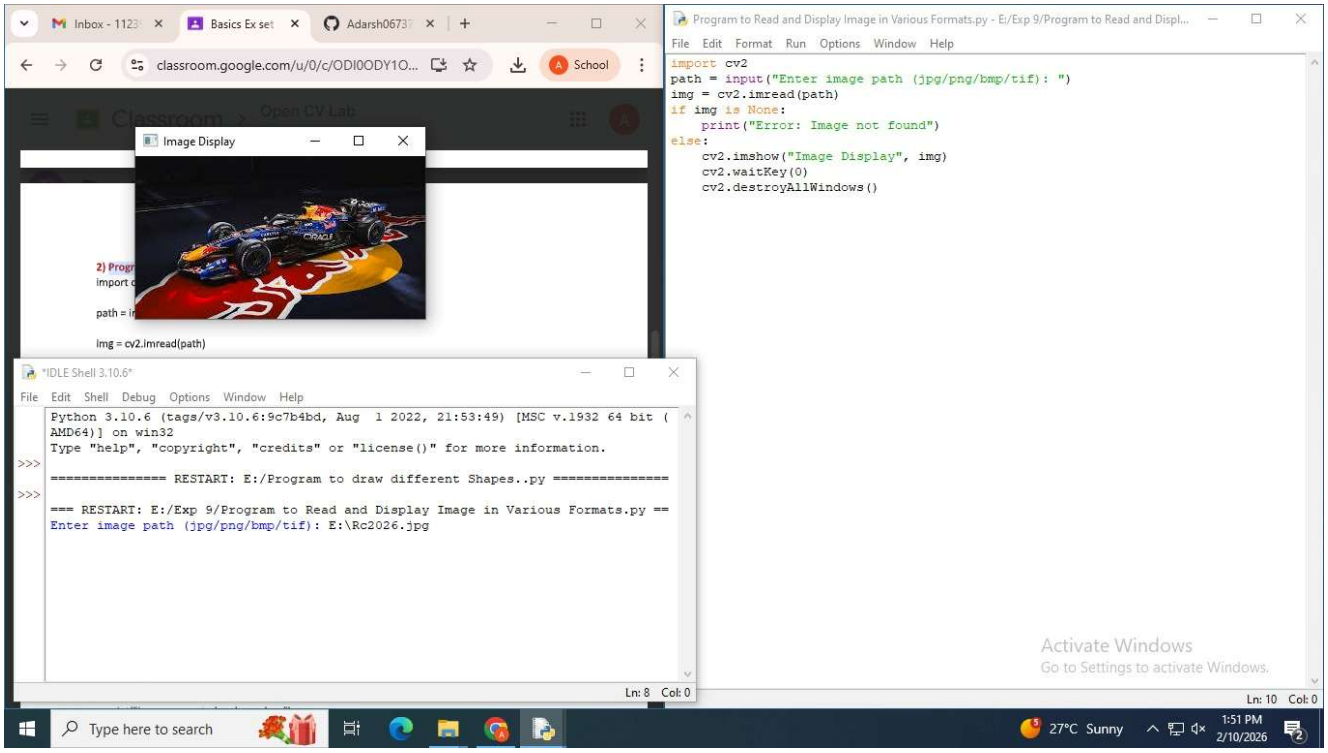
CODE:

```
import cv2

formats = ['image.jpg', 'image.png', 'image.bmp', 'image.tiff']

for filename in formats:
    img = cv2.imread(filename)
    if img is not None:
        cv2.imshow(f'Format: {filename}', img)
        cv2.waitKey(0)
    else:
        print(f"Could not read: {filename}")

cv2.destroyAllWindows()
```

RESULT:

Images in JPG, PNG, BMP, and TIFF formats were successfully read and displayed using OpenCV.

Output :


Program to Read and Display Image in Various Formats.py - E:/Exp 9/Program to Read and Displ...
File Edit Format Run Options Window Help
import cv2
path = input("Enter image path (jpg/png/bmp/tif): ")
img = cv2.imread(path)
if img is None:
print("Error: Image not found")
else:
cv2.imshow("Image Display", img)
cv2.waitKey(0)
cv2.destroyAllWindows()
Image Display
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Program to draw different Shapes..py ==========
>>>
=== RESTART: E:/Exp 9/Program to Read and Display Image in Various Formats.py ==
Enter image path (jpg/png/bmp/tif): E:\Rc2026.jpg
Ln: 8 Col: 0
Ln: 10 Col: 0
Activate Windows
Go to Settings to activate Windows.
Type here to search
27°C Sunny
1:51 PM
2/10/2026

Ex. No.: 08 **CONVERT IMAGE TO VARIOUS FORMATS**

Date: 09.02.2026

AIM:

To read an image and save (convert) it into multiple file formats including JPG, PNG, BMP, and TIFF.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the source image using cv2.imread('image.jpg').
3. Use cv2.imwrite('output.png', img) to save the image in PNG format.
4. Use cv2.imwrite('output.bmp', img) to save in BMP format.
5. Use cv2.imwrite('output.tiff', img) to save in TIFF format.
6. Print a confirmation message after each save to verify success.

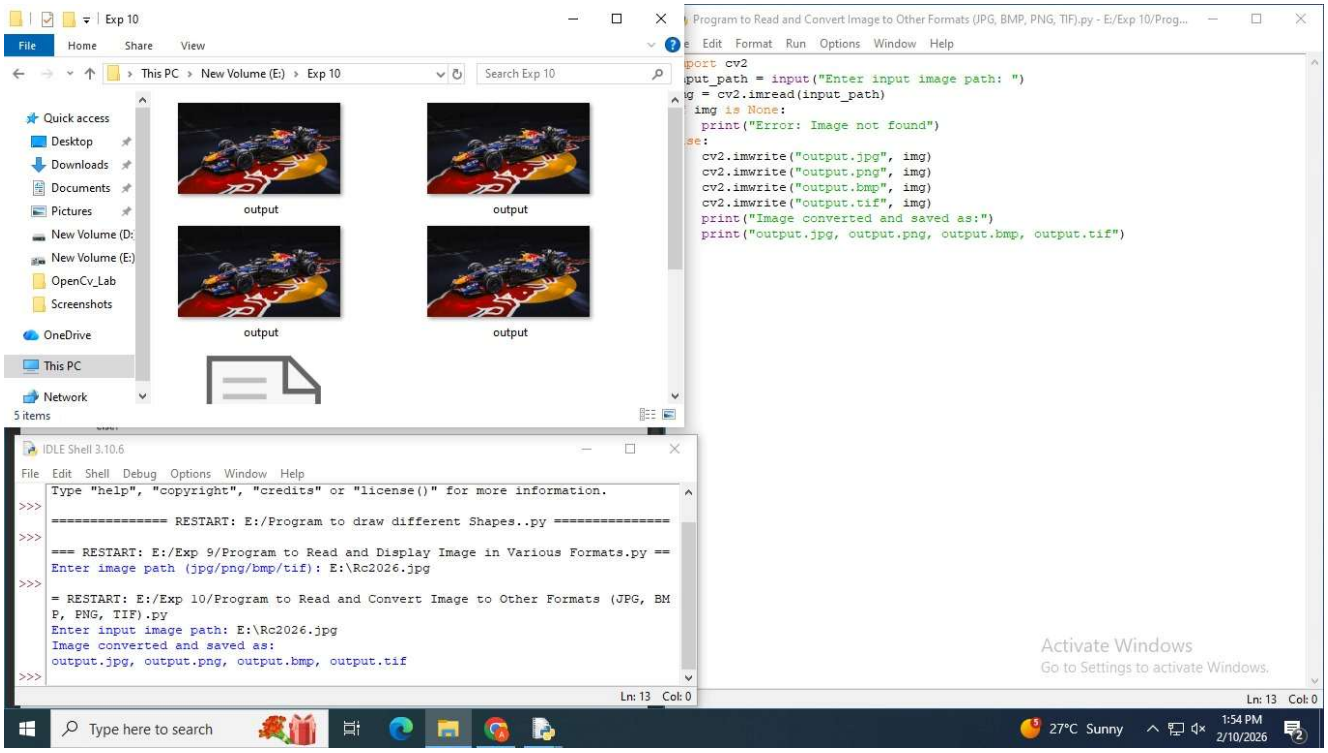
CODE:

```
import cv2

img = cv2.imread('image.jpg')

cv2.imwrite('output.png', img)
print("Saved as PNG")

cv2.imwrite('output.bmp', img)
print("Saved as BMP")

cv2.imwrite('output.tiff', img)
print("Saved as TIFF")

cv2.imwrite('output_copy.jpg', img)
print("Saved as JPG")

print("All formats saved successfully.")
```

RESULT:

The image was successfully converted and saved in JPG, PNG, BMP, and TIFF formats.

Output :


Exp 10
File Home Share View
This PC › New Volume (E:) › Exp 10
Search Exp 10
Quick access
Desktop
Downloads
Documents
Pictures
New Volume (D:
New Volume (E:)
OpenCv_Lab
Screenshots
OneDrive
This PC
Network
5 items
output
output
output
output
Program to Read and Convert Image to Other Formats (JPG, BMP, PNG, TIF).py - E:/Exp 10/Prog...
Edit Format Run Options Window Help
port cv2
put_path = input("Enter input image path: ")
g = cv2.imread(input_path)
img is None:
print("Error: Image not found")
se:
cv2.imwrite("output.jpg", img)
cv2.imwrite("output.png", img)
cv2.imwrite("output.bmp", img)
cv2.imwrite("output.tif", img)
print("Image converted and saved as:")
print("output.jpg, output.png, output.bmp, output.tif")
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Program to draw different Shapes..py ==========
>>>
=== RESTART: E:/Exp 9/Program to Read and Display Image in Various Formats.py ==
Enter image path (jpg/png/bmp/tif): E:\Rc2026.jpg
>>>
= RESTART: E:/Exp 10/Program to Read and Convert Image to Other Formats (JPG, BM
P, PNG, TIF).py
Enter input image path: E:\Rc2026.jpg
Image converted and saved as:
output.jpg, output.png, output.bmp, output.tif
>>>
Ln: 13 Col: 0
Activate Windows
Go to Settings to activate Windows.
Ln: 13 Col: 0
Type here to search
27°C Sunny
1:54 PM
2/10/2026

Ex. No.: 09 **ROTATE IMAGE AT VARIOUS ANGLES**

Date: 09.02.2026

AIM:

To rotate an image at multiple angles (45°, 90°, 180°) using OpenCV's affine transformation.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image and get its dimensions using img.shape[:2].
3. Compute the center of the image for rotation.
4. Use cv2.getRotationMatrix2D(center, angle, scale) to generate the rotation matrix.
5. Apply the rotation using cv2.warpAffine(img, M, (width, height)).
6. Repeat for angles 45°, 90°, and 180°, displaying each result.

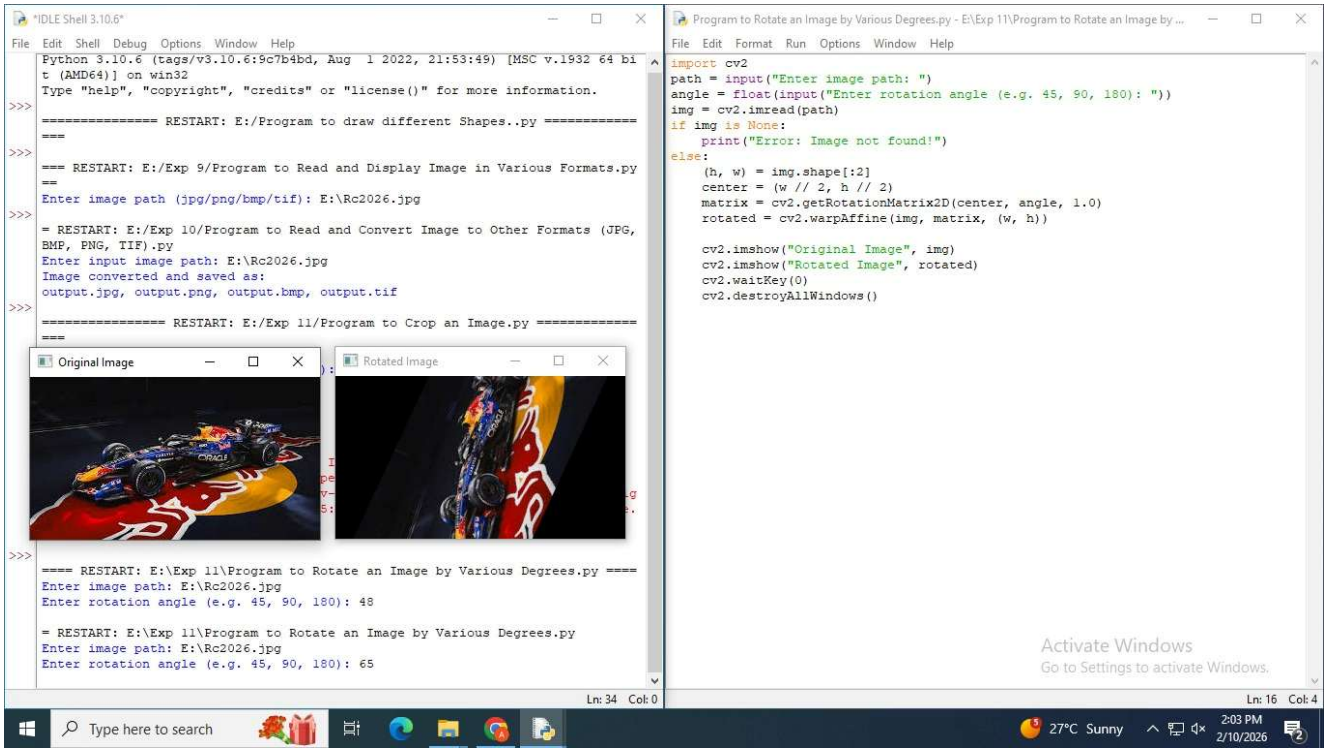
CODE:

```
import cv2

img = cv2.imread('image.jpg')
h, w = img.shape[:2]
center = (w // 2, h // 2)

for angle in [45, 90, 180]:
    M = cv2.getRotationMatrix2D(center, angle, 1.0)
    rotated = cv2.warpAffine(img, M, (w, h))
    cv2.imshow(f'Rotated {angle} degrees', rotated)
    cv2.waitKey(0)

cv2.destroyAllWindows()
```

RESULT:

The image was successfully rotated at 45°, 90°, and 180° and displayed.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Program to draw different Shapes..py ==========
>>>
=== RESTART: E:/Exp 9/Program to Read and Display Image in Various Formats.py ==
Enter image path (jpg/png/bmp/tif): E:\Rc2026.jpg
>>>
= RESTART: E:/Exp 10/Program to Read and Convert Image to Other Formats (JPG, BMP, PNG, TIF).py
Enter input image path: E:\Rc2026.jpg
Image converted and saved as:
output.jpg, output.png, output.bmp, output.tif
>>>
========== RESTART: E:/Exp 11/Program to Crop an Image.py ==========
Original Image
Rotated Image
>>>
==== RESTART: E:\Exp 11\Program to Rotate an Image by Various Degrees.py ====
Enter image path: E:\Rc2026.jpg
Enter rotation angle (e.g. 45, 90, 180): 48
= RESTART: E:\Exp 11\Program to Rotate an Image by Various Degrees.py
Enter image path: E:\Rc2026.jpg
Enter rotation angle (e.g. 45, 90, 180): 65
Ln: 34 Col: 0
Program to Rotate an Image by Various Degrees.py - E:\Exp 11\Program to Rotate an Image by ...
File Edit Format Run Options Window Help
import cv2
path = input("Enter image path: ")
angle = float(input("Enter rotation angle (e.g. 45, 90, 180): "))
img = cv2.imread(path)
if img is None:
print("Error: Image not found!")
else:
(h, w) = img.shape[:2]
center = (w // 2, h // 2)
matrix = cv2.getRotationMatrix2D(center, angle, 1.0)
rotated = cv2.warpAffine(img, matrix, (w, h))
cv2.imshow("Original Image", img)
cv2.imshow("Rotated Image", rotated)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 16 Col: 4
Type here to search
27°C Sunny
2:03 PM
2/10/2026

Ex. No.: 10 **CROP AN IMAGE**

Date: 09.02.2026

AIM:

To crop a specific region of interest (ROI) from an image using NumPy array slicing.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image using cv2.imread('image.jpg').
3. Identify the desired region by specifying pixel coordinates: start and end values for rows (y) and columns (x).
4. Crop the region using NumPy slicing syntax: cropped = img[y1:y2, x1:x2].
5. Display the cropped region using cv2.imshow().
6. Wait for a key press and close all windows.

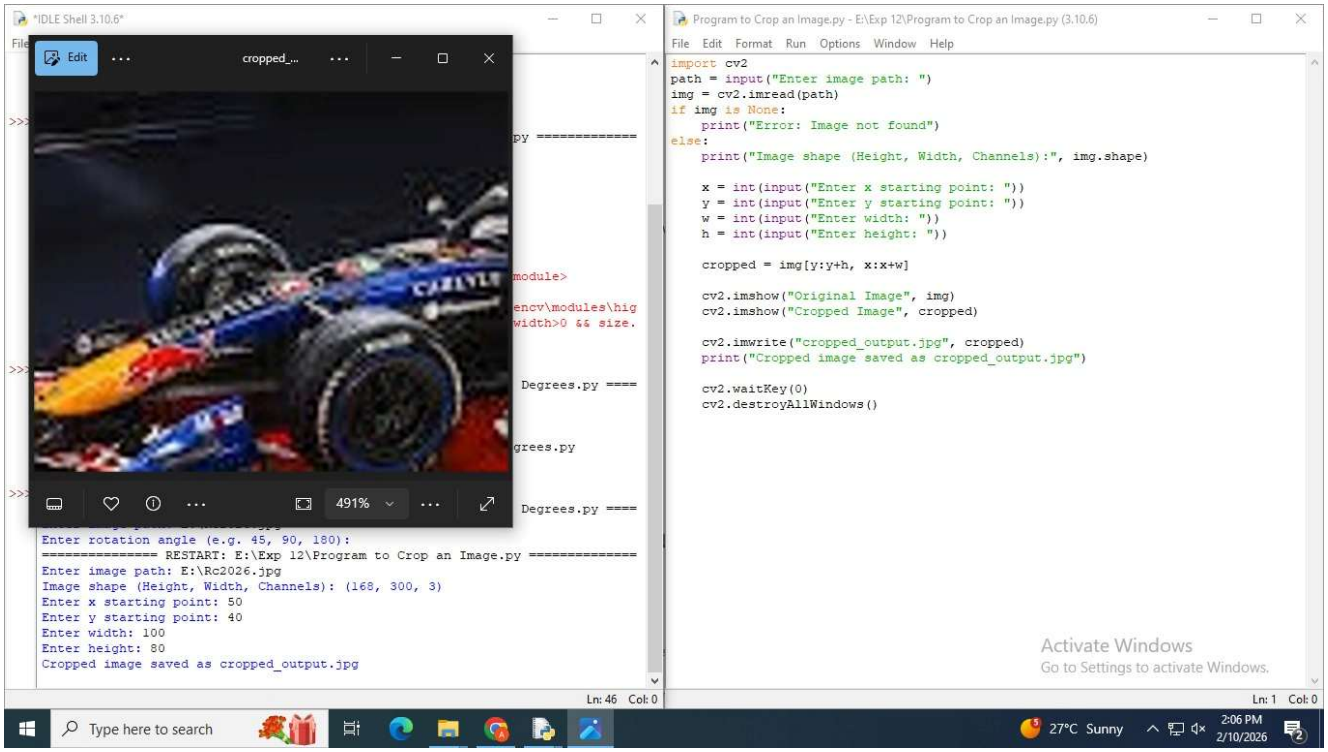
CODE:

```
import cv2

img = cv2.imread('image.jpg')

# Crop region: rows 50 to 200, columns 100 to 300
cropped = img[50:200, 100:300]

cv2.imshow('Original Image', img)
cv2.imshow('Cropped Region', cropped)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

A specific region of interest was successfully cropped from the image and displayed.

Output :


IDLE Shell 3.10.6
Enter rotation angle (e.g. 45, 90, 180):
========== RESTART: E:\Exp 12\Program to Crop an Image.py ==========
Enter image path: E:\Rc2026.jpg
Image shape (Height, Width, Channels): (168, 300, 3)
Enter x starting point: 50
Enter y starting point: 40
Enter width: 100
Enter height: 80
Cropped image saved as cropped_output.jpg
Ln: 46 Col: 0
Program to Crop an Image.py - E:\Exp 12\Program to Crop an Image.py (3.10.6)
File Edit Format Run Options Window Help
import cv2
path = input("Enter image path: ")
img = cv2.imread(path)
if img is None:
print("Error: Image not found")
else:
print("Image shape (Height, Width, Channels):", img.shape)
x = int(input("Enter x starting point: "))
y = int(input("Enter y starting point: "))
w = int(input("Enter width: "))
h = int(input("Enter height: "))
cropped = img[y:y+h, x:x+w]
cv2.imshow("Original Image", img)
cv2.imshow("Cropped Image", cropped)
cv2.imwrite("cropped_output.jpg", cropped)
print("Cropped image saved as cropped_output.jpg")
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 1 Col: 0
Edit
cropped_...
491%
Type here to search
27°C Sunny
2:06 PM
2/10/2026

Ex. No.: 11 **IMAGE FLIPPING**

Date: 17.02.2026

AIM:

To flip an image horizontally, vertically, and in both directions using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image using cv2.imread('image.jpg').
3. Use cv2.flip(img, 1) for horizontal flip (left-right).
4. Use cv2.flip(img, 0) for vertical flip (top-bottom).
5. Use cv2.flip(img, -1) to flip in both directions simultaneously.
6. Display the original and all three flipped images.

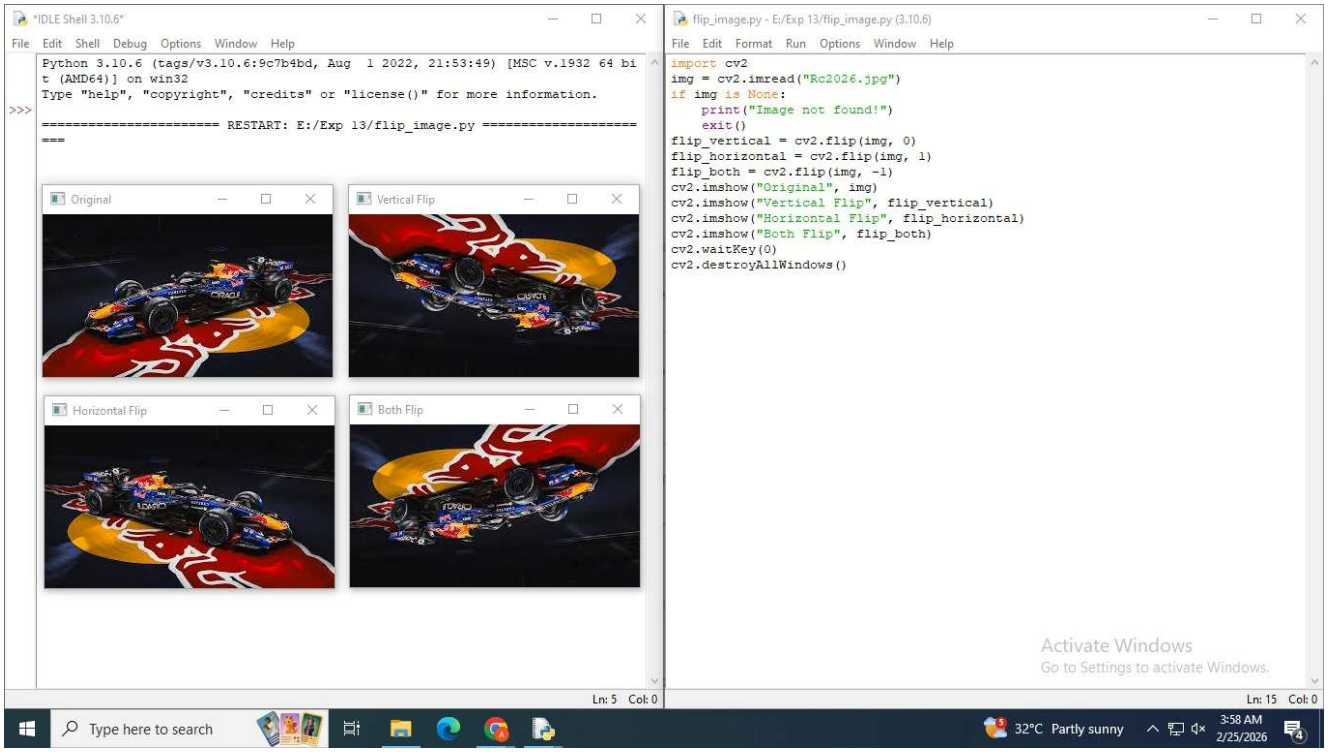
CODE:

```
import cv2

img = cv2.imread('image.jpg')

h_flip = cv2.flip(img, 1)   # Horizontal
v_flip = cv2.flip(img, 0)   # Vertical
both   = cv2.flip(img, -1)  # Both

cv2.imshow('Original', img)
cv2.imshow('Horizontal Flip', h_flip)
cv2.imshow('Vertical Flip', v_flip)
cv2.imshow('Both Directions', both)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The image was successfully flipped horizontally, vertically, and in both directions.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Exp 13/flip_image.py ==========
Original
Vertical Flip
Horizontal Flip
Both Flip
Ln: 5 Col: 0
flip_image.py - E:/Exp 13/flip_image.py (3.10.6)
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
if img is None:
print("Image not found!")
exit()
flip_vertical = cv2.flip(img, 0)
flip_horizontal = cv2.flip(img, 1)
flip_both = cv2.flip(img, -1)
cv2.imshow("Original", img)
cv2.imshow("Vertical Flip", flip_vertical)
cv2.imshow("Horizontal Flip", flip_horizontal)
cv2.imshow("Both Flip", flip_both)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 15 Col: 0
Type here to search
32°C Partly sunny
3:58 AM
2/25/2026

Ex. No.: 12 **IMAGE SMOOTHING AND BLURRING**

Date: 17.02.2026

AIM:

To apply different blurring techniques — Average, Gaussian, and Median blur — to an image using OpenCV.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image using cv2.imread('image.jpg').
3. Apply Average Blur using cv2.blur(img, (5, 5)) with a 5x5 kernel.
4. Apply Gaussian Blur using cv2.GaussianBlur(img, (5, 5), 0).
5. Apply Median Blur using cv2.medianBlur(img, 5).
6. Display all results side by side for comparison.

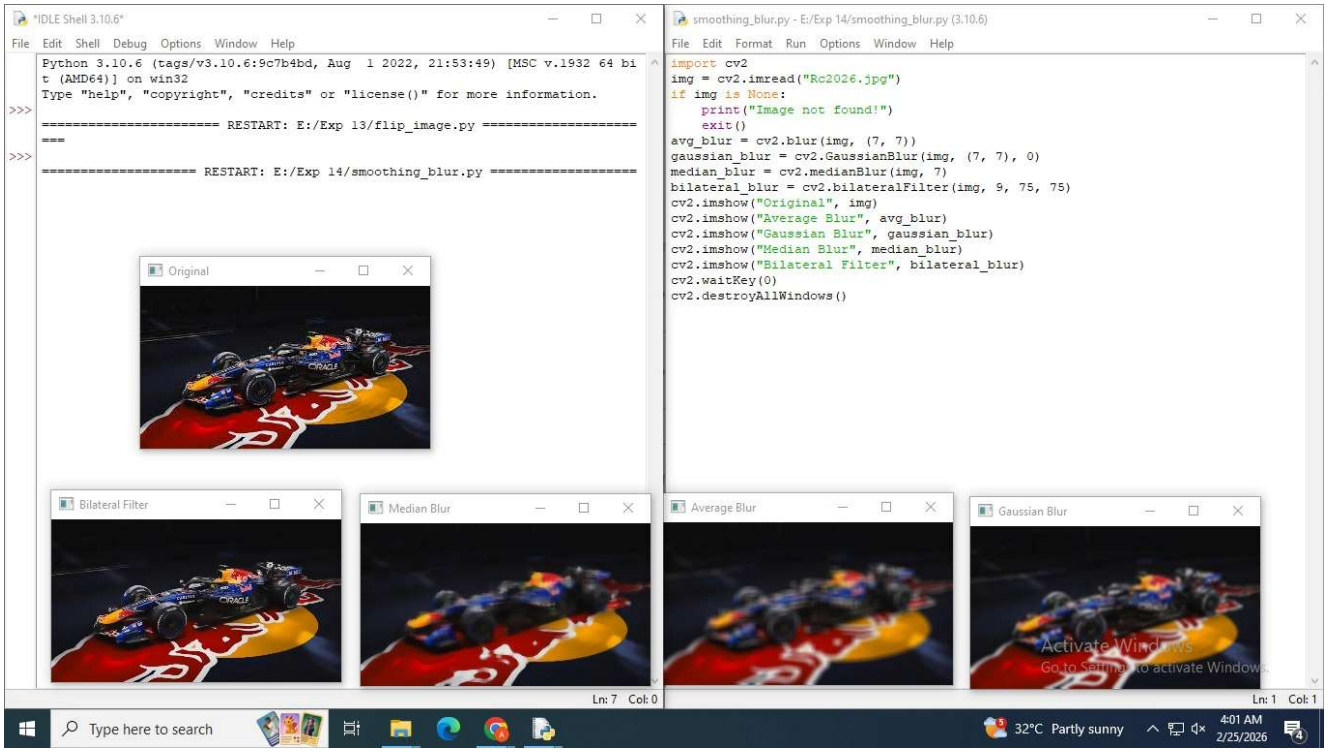
CODE:

```
import cv2

img = cv2.imread('image.jpg')

avg_blur   = cv2.blur(img, (5, 5))
gauss_blur = cv2.GaussianBlur(img, (5, 5), 0)
med_blur   = cv2.medianBlur(img, 5)

cv2.imshow('Original', img)
cv2.imshow('Average Blur', avg_blur)
cv2.imshow('Gaussian Blur', gauss_blur)
cv2.imshow('Median Blur', med_blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Average, Gaussian, and Median blurring techniques were applied successfully to the image.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Exp 13/flip_image.py ==========
>>>
========== RESTART: E:/Exp 14/smoothing_blur.py ==========
smoothing_blur.py - E:/Exp 14/smoothing_blur.py (3.10.6)
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
if img is None:
print("Image not found!")
exit()
avg_blur = cv2.blur(img, (7, 7))
gaussian_blur = cv2.GaussianBlur(img, (7, 7), 0)
median_blur = cv2.medianBlur(img, 7)
bilateral_blur = cv2.bilateralFilter(img, 9, 75, 75)
cv2.imshow("Original", img)
cv2.imshow("Average Blur", avg_blur)
cv2.imshow("Gaussian Blur", gaussian_blur)
cv2.imshow("Median Blur", median_blur)
cv2.imshow("Bilateral Filter", bilateral_blur)
cv2.waitKey(0)
cv2.destroyAllWindows()
Original
Bilateral Filter
Median Blur
Average Blur
Gaussian Blur
Ln: 7 Col: 0
Ln: 1 Col: 1
Type here to search
32°C Partly sunny
4:01 AM
2/25/2026

Ex. No.: 13 **HISTOGRAM ANALYSIS**

Date: 17.02.2026

AIM:

To compute and plot the pixel intensity histogram of a grayscale image using OpenCV and Matplotlib.

PROCEDURE:

1. Import cv2 and matplotlib.pyplot as plt.
2. Load the image and convert it to grayscale using cv2.cvtColor().
3. Compute the histogram using cv2.calcHist([gray], [0], None, [256], [0, 256]).
4. Plot the histogram using plt.plot(hist).
5. Add labels and title to the plot for clarity.
6. Call plt.show() to display the histogram graph.

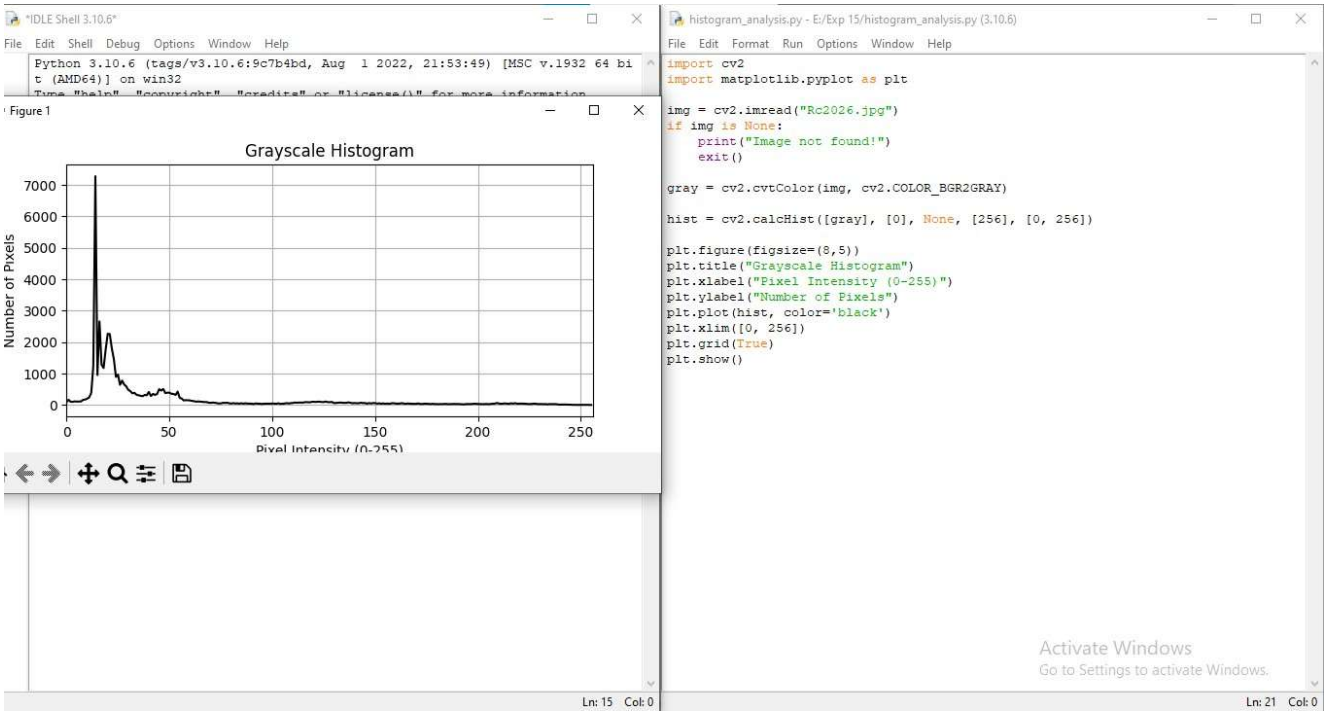
CODE:

```
import cv2
import matplotlib.pyplot as plt

img  = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

hist = cv2.calcHist([gray], [0], None, [256], [0, 256])

plt.figure()
plt.plot(hist, color='black')
plt.title('Grayscale Histogram')
plt.xlabel('Pixel Intensity (0-255)')
plt.ylabel('Frequency')
plt.xlim([0, 256])
plt.show()
```

RESULT:

The histogram of the grayscale image was computed and displayed, showing the distribution of pixel intensities.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Figure 1
Grayscale Histogram
Number of Pixels
Pixel Intensity (0-255)
Ln: 15 Col: 0
histogram_analysis.py - E:/Exp 15/histogram_analysis.py (3.10.6)
File Edit Format Run Options Window Help
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("Rc2026.jpg")
if img is None:
print("Image not found!")
exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
hist = cv2.calcHist([gray], [0], None, [256], [0, 256])
plt.figure(figsize=(8,5))
plt.title("Grayscale Histogram")
plt.xlabel("Pixel Intensity (0-255)")
plt.ylabel("Number of Pixels")
plt.plot(hist, color='black')
plt.xlim([0, 256])
plt.grid(True)
plt.show()
Activate Windows
Go to Settings to activate Windows.
Ln: 21 Col: 0

Ex. No.: 14 **EDGE DETECTION**

Date: 02.03.2026

AIM:

To detect edges in an image using the Canny edge detection algorithm.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Load the image and convert it to grayscale using cv2.cvtColor().
3. Apply cv2.Canny(gray, threshold1, threshold2) with lower threshold 100 and upper threshold 200.
4. The Canny function automatically applies Gaussian smoothing, gradient computation, and hysteresis thresholding.
5. Display the resulting edge-detected image using cv2.imshow().
6. Wait for a key press and close all windows.

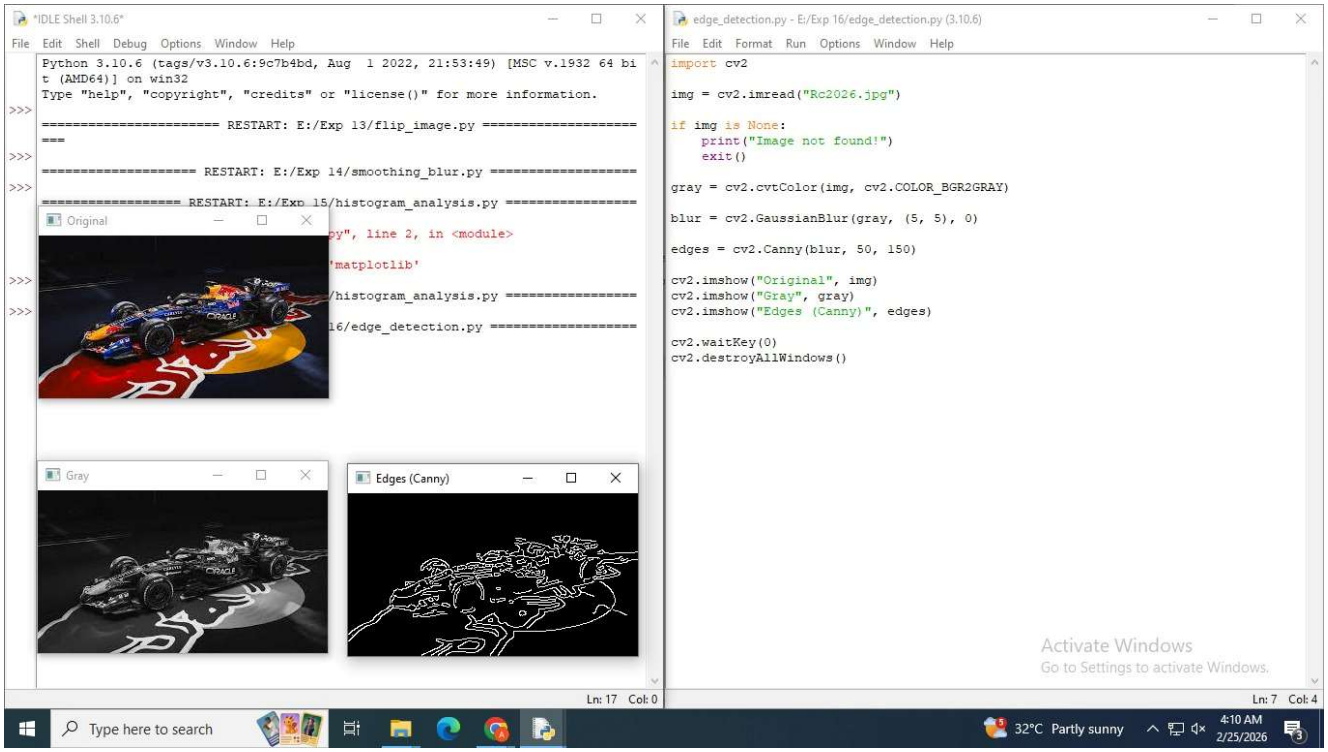
CODE:

```
import cv2

img  = cv2.imread('image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

edges = cv2.Canny(gray, 100, 200)

cv2.imshow('Original', img)
cv2.imshow('Edge Detection (Canny)', edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Edges in the image were successfully detected using the Canny edge detector and displayed.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:/Exp 13/flip_image.py ==========
>>>
========== RESTART: E:/Exp 14/smoothing_blur.py ==========
>>>
========== RESTART: E:/Exp 15/histogram_analysis.py ==========
Original
Gray
Edges (Canny)
Ln: 17 Col: 0
edge_detection.py - E:/Exp 16/edge_detection.py (3.10.6)
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg")
if img is None:
print("Image not found!")
exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
blur = cv2.GaussianBlur(gray, (5, 5), 0)
edges = cv2.Canny(blur, 50, 150)
cv2.imshow("Original", img)
cv2.imshow("Gray", gray)
cv2.imshow("Edges (Canny)", edges)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 7 Col: 4
Type here to search
32°C Partly sunny
4:10 AM
2/25/2026

Ex. No.: 15 **BINARY IMAGE PROCESSING**

Date: 02.03.2026

AIM:

To apply binary thresholding and demonstrate erosion, dilation, opening, and closing on a grayscale image.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode using cv2.imread('image.jpg', 0).
3. Apply binary thresholding using cv2.threshold(img, 127, 255, cv2.THRESH_BINARY).
4. Create a 5x5 kernel using np.ones((5, 5), np.uint8).
5. Apply erosion using cv2.erode(), dilation using cv2.dilate(), opening and closing using cv2.morphologyEx().
6. Display all results for comparison.

CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel = np.ones((5, 5), np.uint8)

eroded  = cv2.erode(binary, kernel, iterations=1)
dilated = cv2.dilate(binary, kernel, iterations=1)
opened  = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
closed  = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)

cv2.imshow('Binary', binary)
cv2.imshow('Eroded', eroded)
cv2.imshow('Dilated', dilated)
cv2.imshow('Opened', opened)
cv2.imshow('Closed', closed)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Binary thresholding, erosion, dilation, opening, and closing were all applied and demonstrated successfully.

Output :

PythonProject2
1c.py
2.a.py
1a.py
import cv2
import numpy as np
img = cv2.imread('bmw.jpg',
Eroded
Dilated
Opened
Binary
Run
1.b
C:\Users\sarva\PycharmProjects
PythonProject2 > 1.b.py
4:22 CRLF UTF-8 4 spaces Python 3.14 (PythonProject2)

Ex. No.: 16 **EROSION**

Date: 02.03.2026

AIM:

To apply the erosion morphological operation to a binary image to shrink foreground objects.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode using cv2.imread('image.jpg', 0).
3. Convert to binary using cv2.threshold().
4. Define a 5x5 kernel using np.ones((5, 5), np.uint8).
5. Apply erosion using cv2.erode(binary, kernel, iterations=1).
6. Display the original binary image and the eroded result.

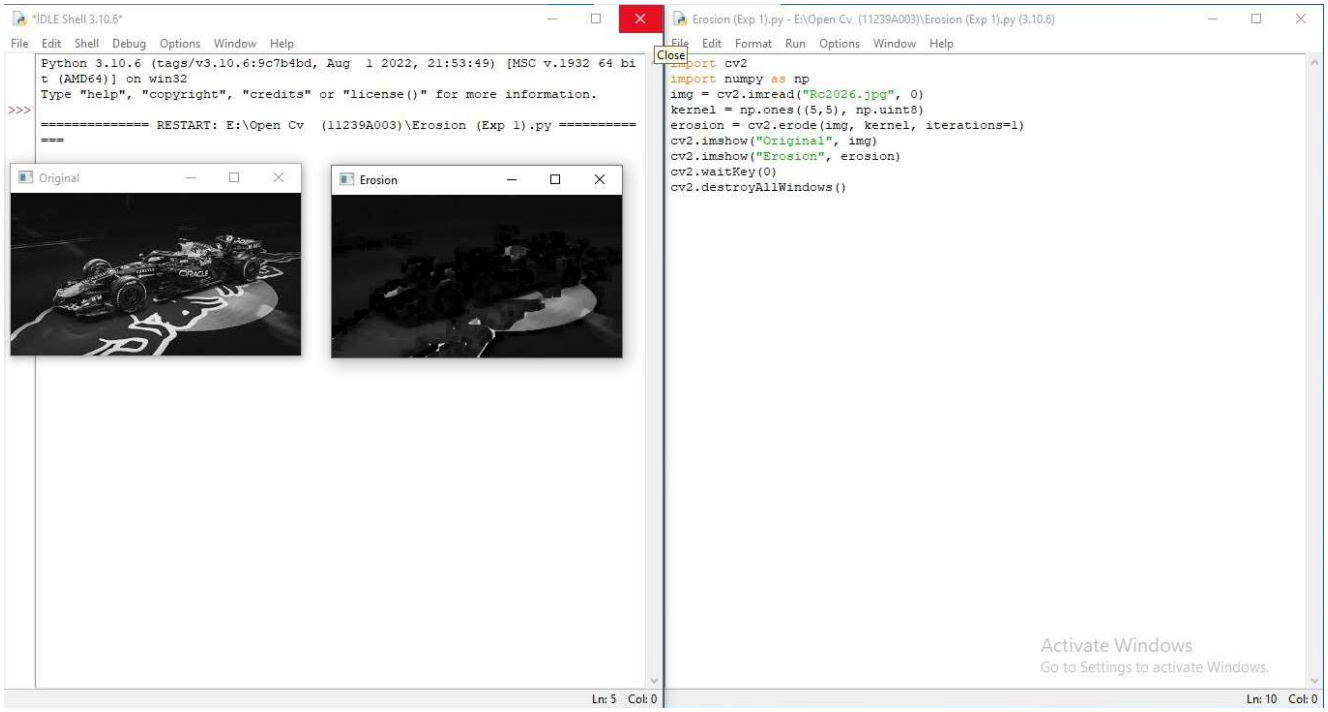
CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel  = np.ones((5, 5), np.uint8)
eroded  = cv2.erode(binary, kernel, iterations=1)

cv2.imshow('Original Binary', binary)
cv2.imshow('Eroded', eroded)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Erosion was applied successfully, shrinking foreground objects in the binary image.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: E:\Open Cv (11239A003)\Erosion (Exp 1).py ==========
Original
Erosion
Ln: 5 Col: 0
Erosion (Exp 1).py - E:\Open Cv (11239A003)\Erosion (Exp 1).py (3.10.6)
File Edit Format Run Options Window Help
Close
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((5,5), np.uint8)
erosion = cv2.erode(img, kernel, iterations=1)
cv2.imshow("Original", img)
cv2.imshow("Erosion", erosion)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 10 Col: 0

Ex. No.: 17 **DILATION**

Date: 02.03.2026

AIM:

To apply the dilation morphological operation to a binary image to expand foreground objects.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode using cv2.imread('image.jpg', 0).
3. Convert to binary using cv2.threshold().
4. Define a 5x5 kernel using np.ones((5, 5), np.uint8).
5. Apply dilation using cv2.dilate(binary, kernel, iterations=1).
6. Display the original binary image and the dilated result.

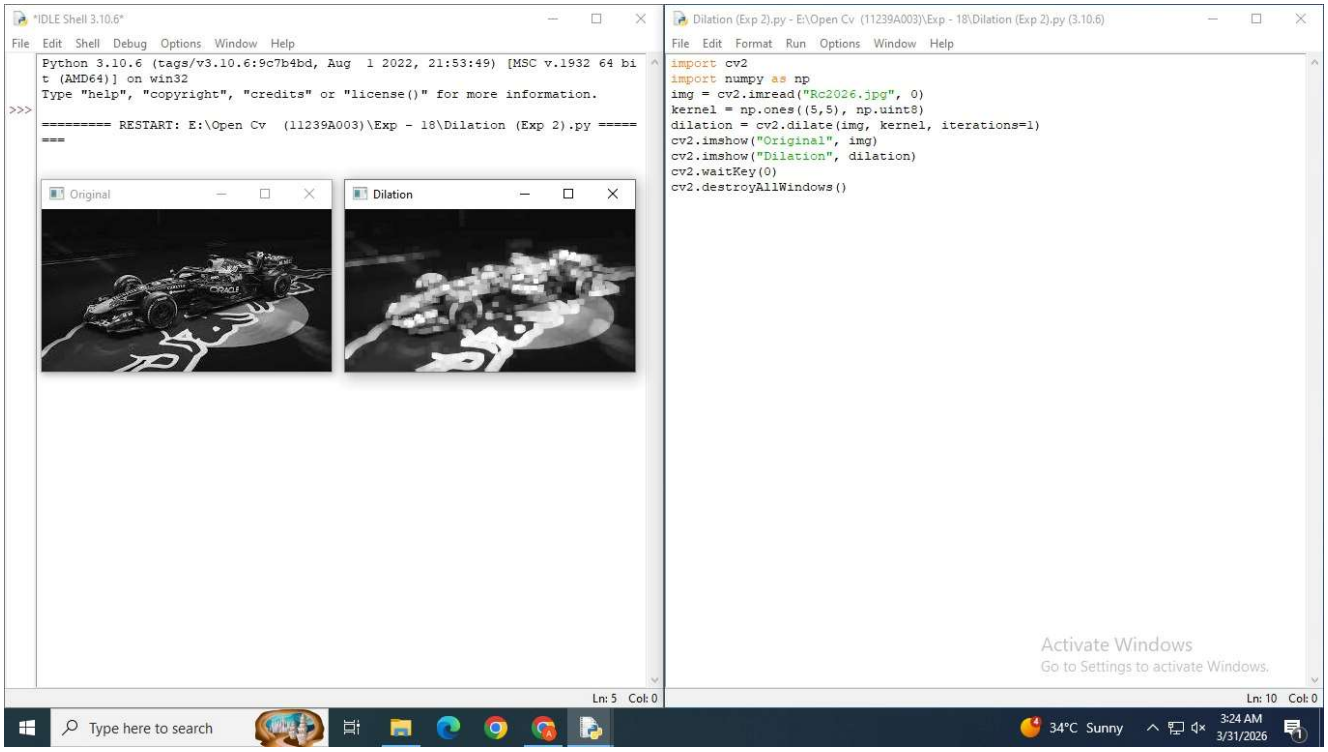
CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel  = np.ones((5, 5), np.uint8)
dilated = cv2.dilate(binary, kernel, iterations=1)

cv2.imshow('Original Binary', binary)
cv2.imshow('Dilated', dilated)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Dilation was applied successfully, expanding foreground objects in the binary image.

Output :


IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========= RESTART: E:\Open Cv (11239A003)\Exp - 18\Dilation (Exp 2).py =========
Original
Dilation
Ln: 5 Col: 0
Dilation (Exp 2).py - E:\Open Cv (11239A003)\Exp - 18\Dilation (Exp 2).py (3.10.6)
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((5,5), np.uint8)
dilation = cv2.dilate(img, kernel, iterations=1)
cv2.imshow("Original", img)
cv2.imshow("Dilation", dilation)
cv2.waitKey(0)
cv2.destroyAllWindows()
Activate Windows
Go to Settings to activate Windows.
Ln: 10 Col: 0
Type here to search
34°C Sunny
3:24 AM
3/31/2026

Ex. No.: 18 **OPENING**

Date: 17.03.2026

AIM:

To apply the morphological opening operation (erosion followed by dilation) to remove small noise from a binary image.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode and convert it to binary using cv2.threshold().
3. Define a 5x5 kernel using np.ones((5, 5), np.uint8).
4. Apply morphological opening using cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel).
5. Display both the original binary and the opened image.
6. Observe the removal of small white noise regions.

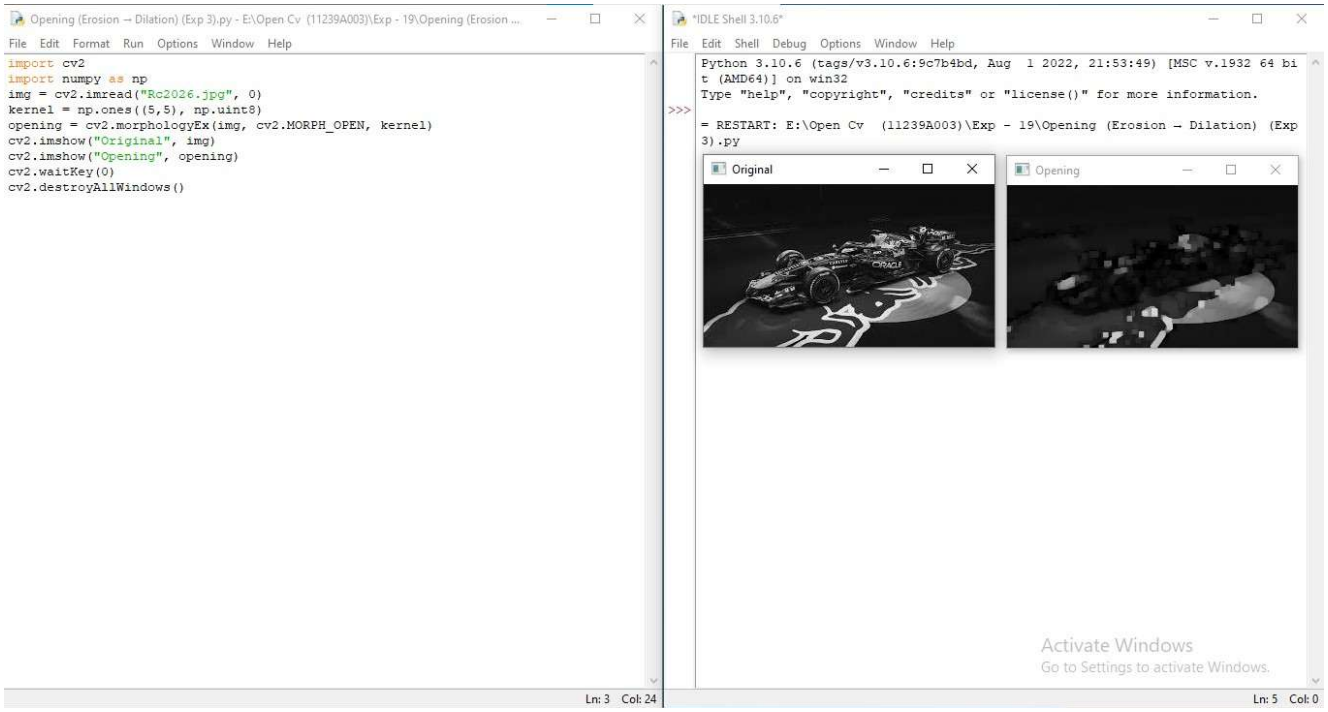
CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel = np.ones((5, 5), np.uint8)
opened = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)

cv2.imshow('Original Binary', binary)
cv2.imshow('Opened', opened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Morphological opening was applied, removing small noise from the binary image while preserving larger objects.

Output :


Opening (Erosion → Dilation) (Exp 3).py - E:\Open Cv (11239A003)\Exp - 19\Opening (Erosion ...
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((5,5), np.uint8)
opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)
cv2.imshow("Original", img)
cv2.imshow("Opening", opening)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 3 Col: 24
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\Open Cv (11239A003)\Exp - 19\Opening (Erosion → Dilation) (Exp 3).py
Original
Opening
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 19 **CLOSING**

Date: 17.03.2026

AIM:

To apply the morphological closing operation (dilation followed by erosion) to fill small holes in foreground objects.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode and apply binary thresholding.
3. Define a 5x5 kernel using np.ones((5, 5), np.uint8).
4. Apply morphological closing using cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel).
5. Display the original binary and the closed image.
6. Observe how small holes inside white regions are filled.

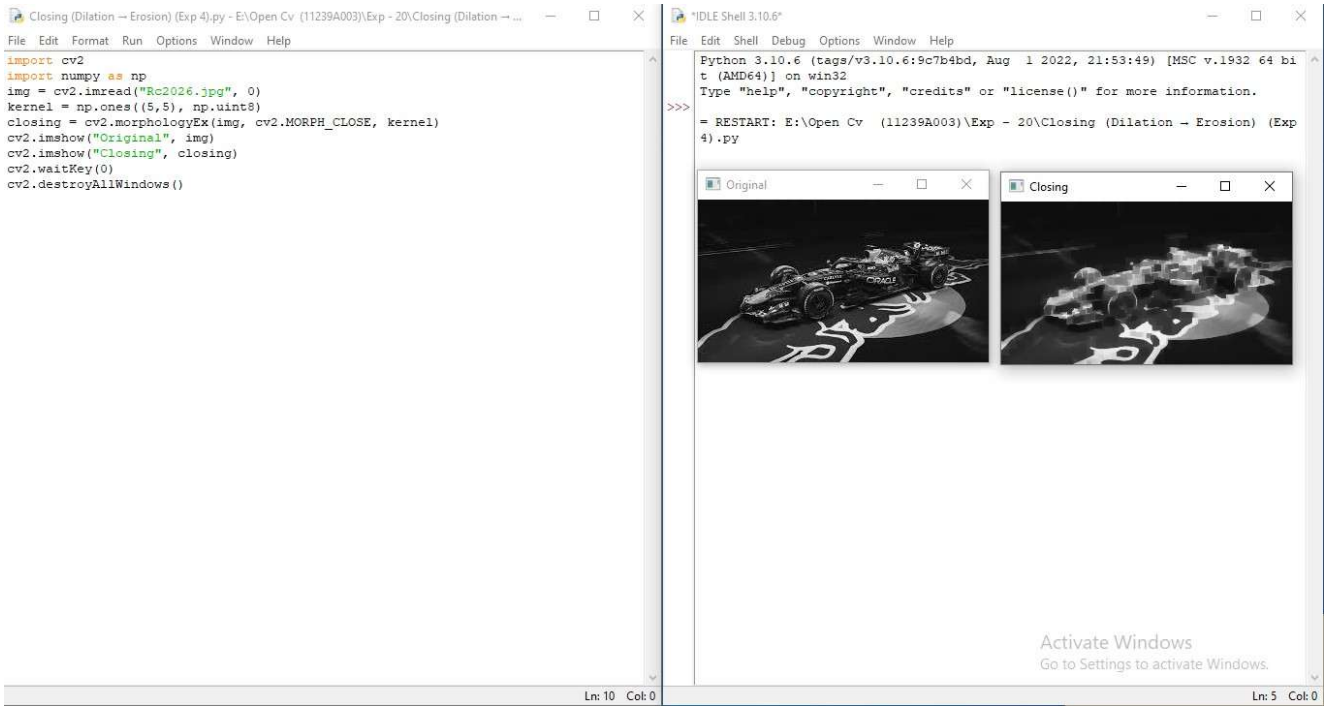
CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel = np.ones((5, 5), np.uint8)
closed = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)

cv2.imshow('Original Binary', binary)
cv2.imshow('Closed', closed)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

Morphological closing was applied successfully, filling small holes within foreground objects.

Output :


Closing (Dilation → Erosion) (Exp 4).py - E:\Open Cv (11239A003)\Exp - 20\Closing (Dilation → ...
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((5,5), np.uint8)
closing = cv2.morphologyEx(img, cv2.MORPH_CLOSE, kernel)
cv2.imshow("Original", img)
cv2.imshow("Closing", closing)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 10 Col: 0
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\Open Cv (11239A003)\Exp - 20\Closing (Dilation → Erosion) (Exp 4).py
Original
Closing
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 20 **MORPHOLOGICAL GRADIENT**

Date: 17.03.2026

AIM:

To compute the morphological gradient of a binary image to highlight object boundaries.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode and apply binary thresholding.
3. Define a 5x5 kernel using np.ones((5, 5), np.uint8).
4. Apply the morphological gradient using cv2.morphologyEx(binary, cv2.MORPH_GRADIENT, kernel).
5. The gradient is computed as the difference between dilation and erosion.
6. Display the result to observe highlighted edges/boundaries.

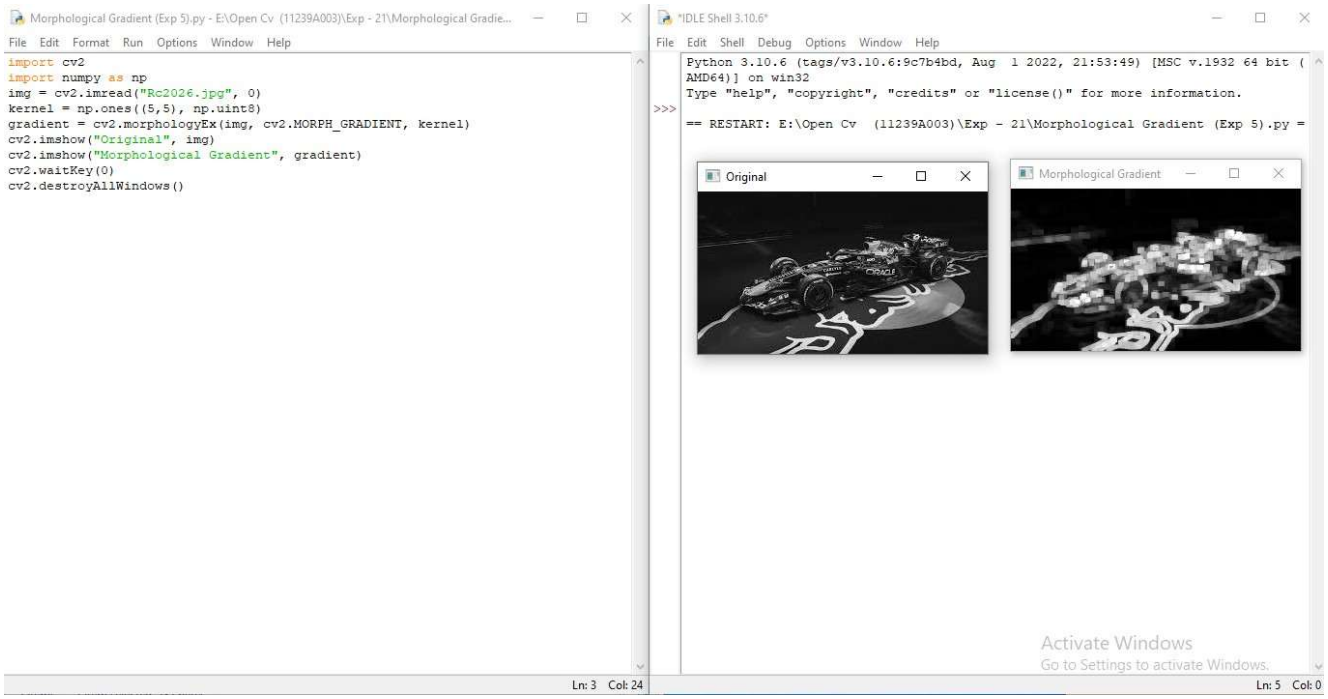
CODE:

```
import cv2
import numpy as np

img = cv2.imread('image.jpg', 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

kernel   = np.ones((5, 5), np.uint8)
gradient = cv2.morphologyEx(binary, cv2.MORPH_GRADIENT, kernel)

cv2.imshow('Original Binary', binary)
cv2.imshow('Morphological Gradient', gradient)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The morphological gradient was computed successfully, highlighting the boundaries of objects in the image.

Output :


Morphological Gradient (Exp 5).py - E:\Open Cv (11239A003)\Exp - 21\Morphological Gradie...
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((5,5), np.uint8)
gradient = cv2.morphologyEx(img, cv2.MORPH_GRADIENT, kernel)
cv2.imshow("Original", img)
cv2.imshow("Morphological Gradient", gradient)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 3 Col: 24
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: E:\Open Cv (11239A003)\Exp - 21\Morphological Gradient (Exp 5).py =
Original
Morphological Gradient
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 21 **TOP HAT TRANSFORMATION**

Date: 17.03.2026

AIM:

To apply the Top Hat transformation to reveal bright features that are smaller than the structuring element.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode using cv2.imread('image.jpg', 0).
3. Define a 9x9 rectangular kernel using np.ones((9, 9), np.uint8).
4. Apply the Top Hat transformation using cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel).
5. Top Hat = Original Image minus Opened Image, revealing small bright regions.
6. Display the original and the Top Hat result.

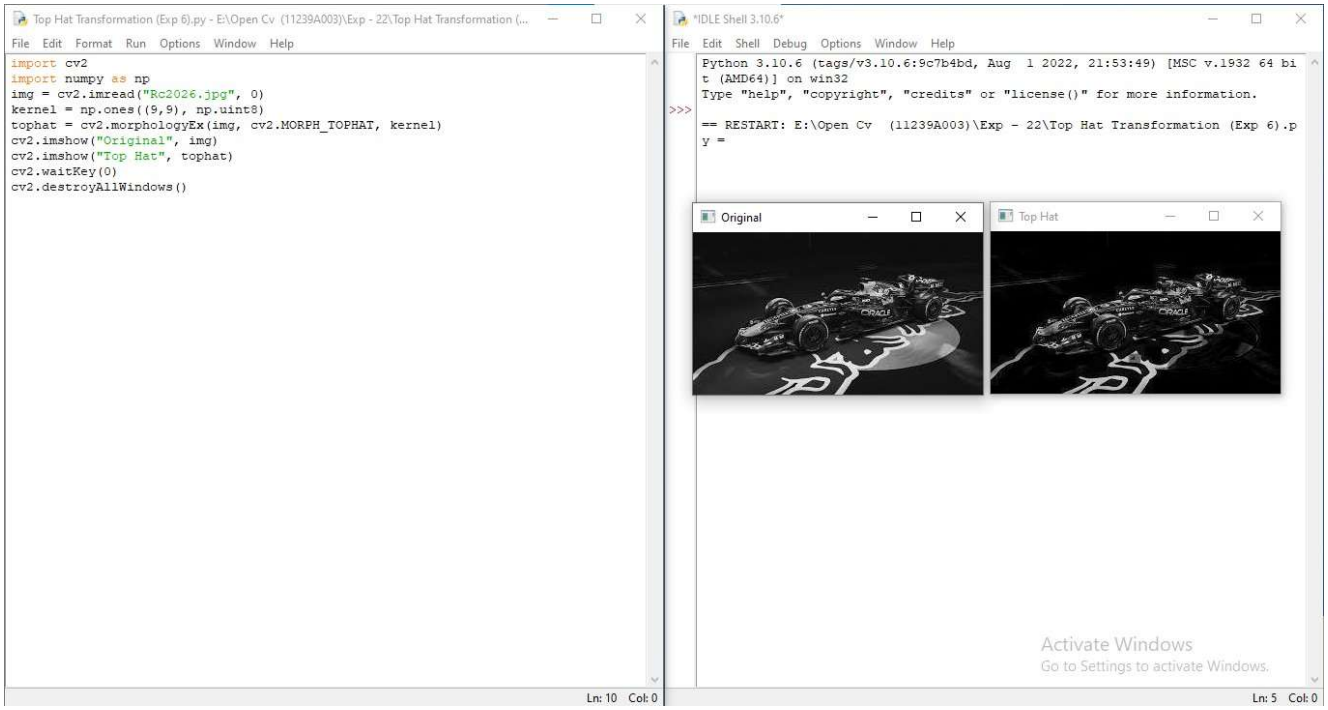
CODE:

```
import cv2
import numpy as np

img    = cv2.imread('image.jpg', 0)
kernel = np.ones((9, 9), np.uint8)
tophat = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)

cv2.imshow('Original', img)
cv2.imshow('Top Hat', tophat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The Top Hat transformation was applied successfully, highlighting small bright regions on a dark background.

Output :


Top Hat Transformation (Exp 6).py - E:\Open Cv (11239A003)\Exp - 22\Top Hat Transformation (...
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((9,9), np.uint8)
tophat = cv2.morphologyEx(img, cv2.MORPH_TOPHAT, kernel)
cv2.imshow("Original", img)
cv2.imshow("Top Hat", tophat)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 10 Col: 0
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
== RESTART: E:\Open Cv (11239A003)\Exp - 22\Top Hat Transformation (Exp 6).py =
Original
Top Hat
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 22 **BLACK HAT TRANSFORMATION**

Date: 30.03.2026

AIM:

To apply the Black Hat transformation to reveal dark features that are smaller than the structuring element.

PROCEDURE:

1. Import cv2 and numpy.
2. Load the image in grayscale mode using cv2.imread('image.jpg', 0).
3. Define a 9x9 rectangular kernel using np.ones((9, 9), np.uint8).
4. Apply the Black Hat transformation using cv2.morphologyEx(img, cv2.MORPH_BLACKHAT, kernel).
5. Black Hat = Closed Image minus Original Image, revealing small dark features.
6. Display both the original and the Black Hat result.

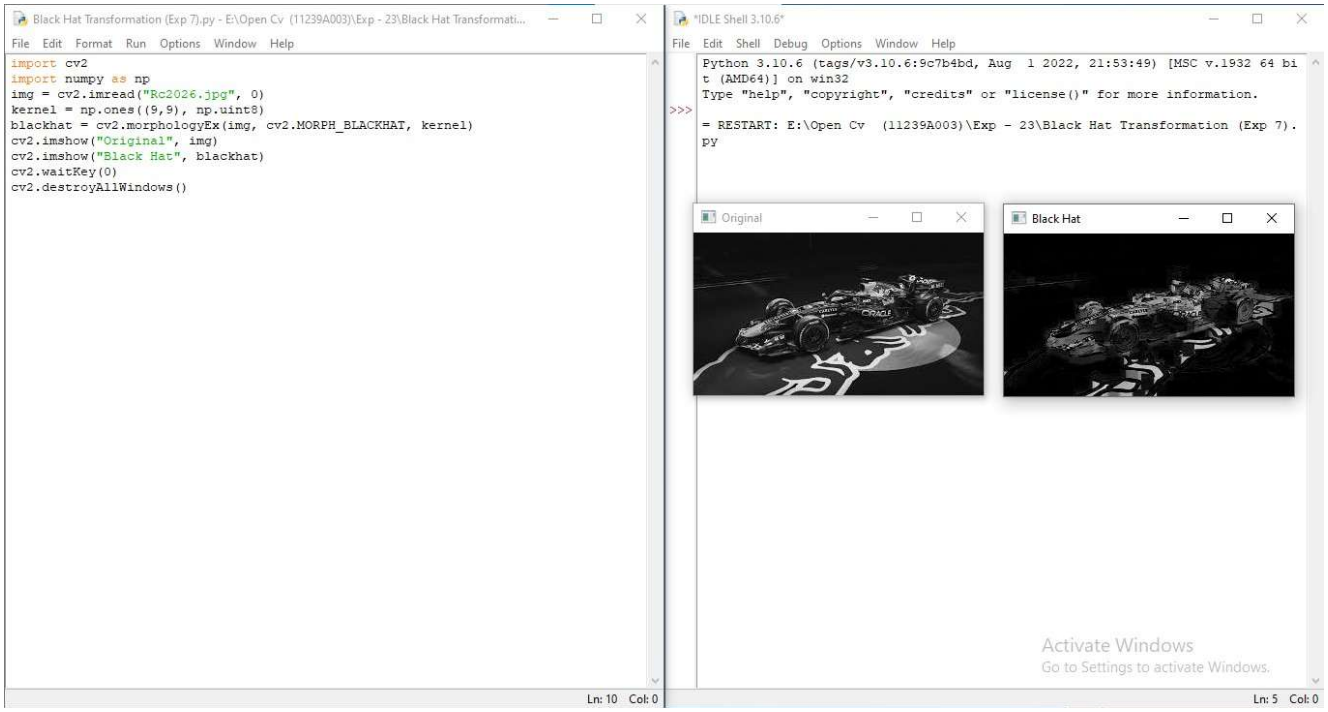
CODE:

```
import cv2
import numpy as np

img     = cv2.imread('image.jpg', 0)
kernel  = np.ones((9, 9), np.uint8)
blackhat = cv2.morphologyEx(img, cv2.MORPH_BLACKHAT, kernel)

cv2.imshow('Original', img)
cv2.imshow('Black Hat', blackhat)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

RESULT:

The Black Hat transformation was applied successfully, highlighting small dark regions on a bright background.

Output :


Black Hat Transformation (Exp 7).py - E:\Open Cv (11239A003)\Exp - 23\Black Hat Transformati...
File Edit Format Run Options Window Help
import cv2
import numpy as np
img = cv2.imread("Rc2026.jpg", 0)
kernel = np.ones((9,9), np.uint8)
blackhat = cv2.morphologyEx(img, cv2.MORPH_BLACKHAT, kernel)
cv2.imshow("Original", img)
cv2.imshow("Black Hat", blackhat)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 10 Col: 0
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\Open Cv (11239A003)\Exp - 23\Black Hat Transformation (Exp 7).py
Original
Black Hat
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 23 **STRUCTURING ELEMENTS**

Date: 30.03.2026

AIM:

To create and display different structuring elements: Rectangular, Elliptical, and Cross-shaped.

PROCEDURE:

1. Import the OpenCV library using import cv2.
2. Use cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5)) to create a rectangular kernel.
3. Use cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5)) to create an elliptical kernel.
4. Use cv2.getStructuringElement(cv2.MORPH_CROSS, (5, 5)) to create a cross-shaped kernel.
5. Print each structuring element to the console to observe its shape.
6. Note that these kernels can be used as parameters in morphological operations.
7.

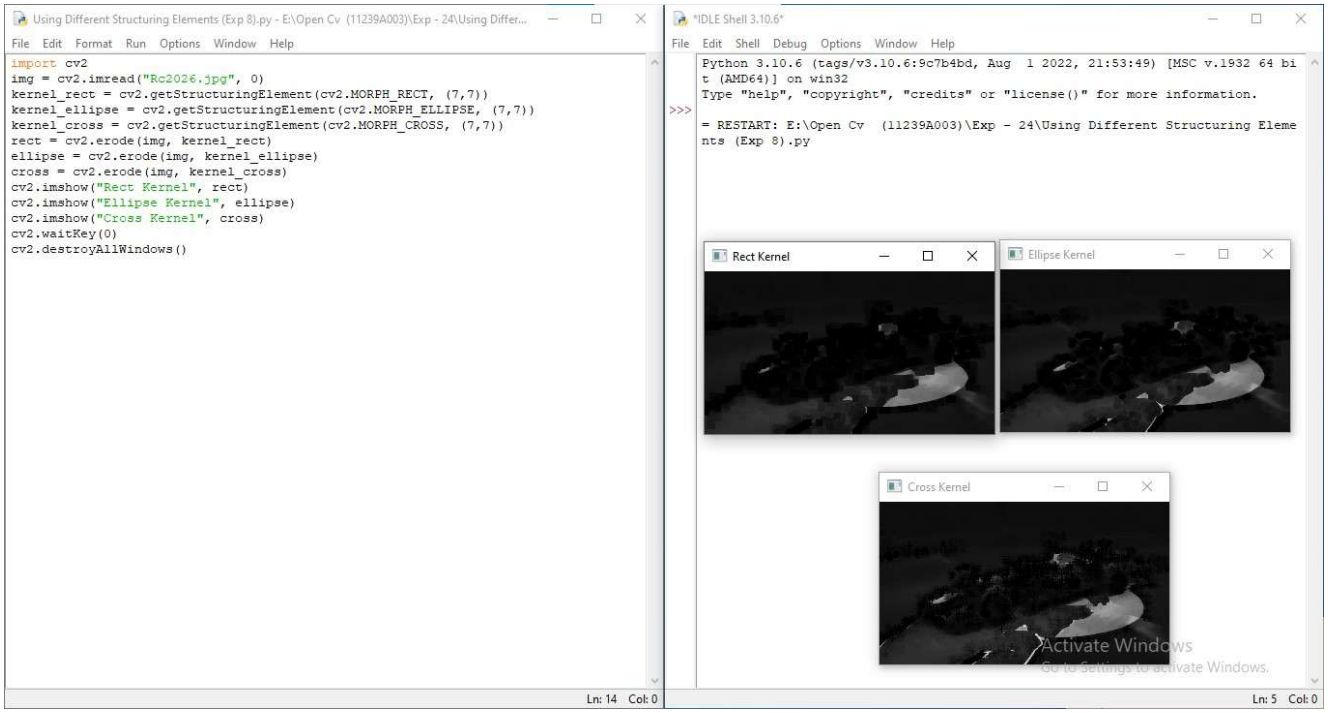
CODE:

```
import cv2

rect_kernel    = cv2.getStructuringElement(cv2.MORPH_RECT,    (5, 5))
ellipse_kernel = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (5, 5))
cross_kernel   = cv2.getStructuringElement(cv2.MORPH_CROSS,   (5, 5))

print("Rectangular Kernel (5x5):")
print(rect_kernel)

print("\nElliptical Kernel (5x5):")
print(ellipse_kernel)

print("\nCross-Shaped Kernel (5x5):")
print(cross_kernel)
```

RESULT:

Rectangular, elliptical, and cross-shaped structuring elements were successfully created and displayed.

Output :


Using Different Structuring Elements (Exp 8).py - E:\Open Cv (11239A003)\Exp - 24\Using Differ...
File Edit Format Run Options Window Help
import cv2
img = cv2.imread("Rc2026.jpg", 0)
kernel_rect = cv2.getStructuringElement(cv2.MORPH_RECT, (7,7))
kernel_ellipse = cv2.getStructuringElement(cv2.MORPH_ELLIPSE, (7,7))
kernel_cross = cv2.getStructuringElement(cv2.MORPH_CROSS, (7,7))
rect = cv2.erode(img, kernel_rect)
ellipse = cv2.erode(img, kernel_ellipse)
cross = cv2.erode(img, kernel_cross)
cv2.imshow("Rect Kernel", rect)
cv2.imshow("Ellipse Kernel", ellipse)
cv2.imshow("Cross Kernel", cross)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 14 Col: 0
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\Open Cv (11239A003)\Exp - 24\Using Different Structuring Elements (Exp 8).py
Rect Kernel
Ellipse Kernel
Cross Kernel
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0

Ex. No.: 24 **BASIC TEXT RECOGNITION FROM IMAGE (OCR USING TESSERACT)**

Date: 30.03.2026

AIM:

To extract printed text from an image using Tesseract OCR and Python's pytesseract library.

PROCEDURE:

1. Install the required libraries: pip install pytesseract and pip install Pillow.
2. Install the Tesseract OCR engine on the system (available at https://github.com/tesseract-ocr/tesseract).
3. Import cv2 and pytesseract.
4. Load the image containing text using cv2.imread().
5. Convert the image to grayscale to improve recognition accuracy.
6. Use pytesseract.image_to_string(gray) to extract the text and print the result.

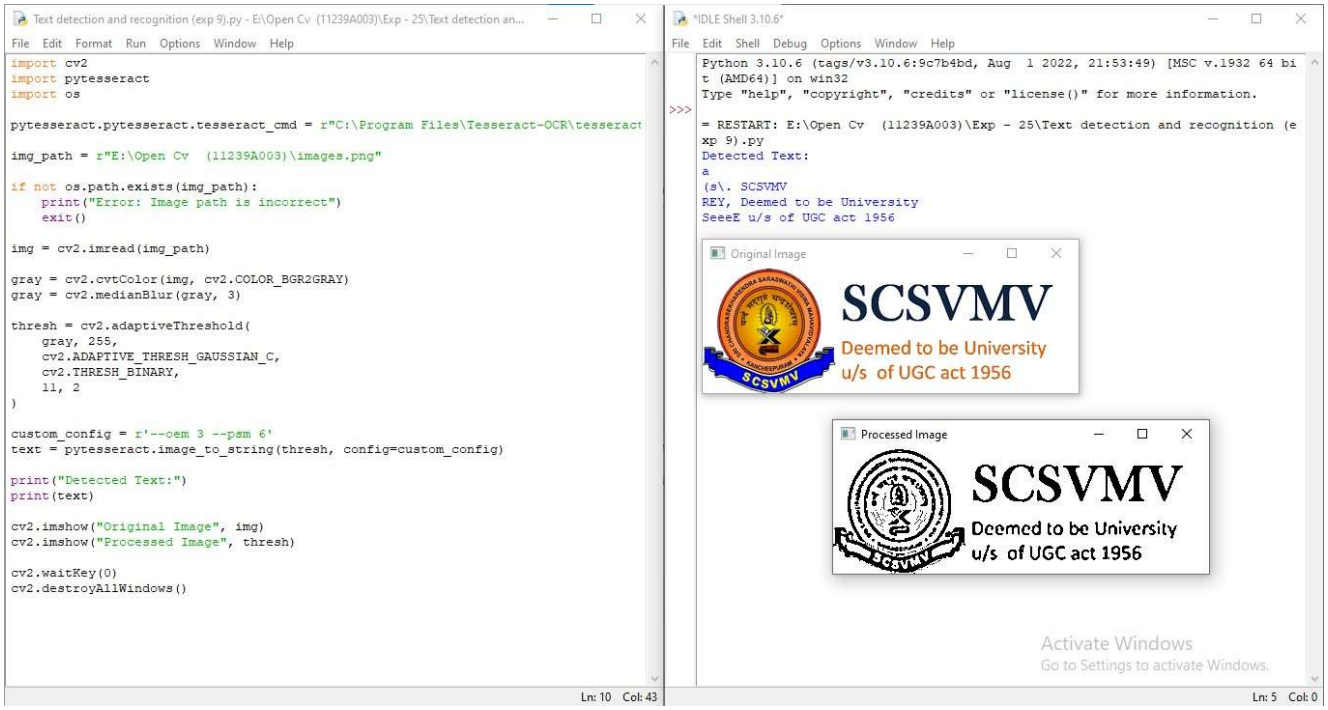
CODE:

```
import cv2
import pytesseract

# On Windows, set the Tesseract path:
# pytesseract.pytesseract.tesseract_cmd = r'C:\Program Files\Tesseract-OCR\tesseract.exe'

img  = cv2.imread('text_image.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

text = pytesseract.image_to_string(gray)

print("Extracted Text:")
print(text)
```

RESULT:

Text was successfully extracted from the image using Tesseract OCR via the pytesseract library.

Output :


Text detection and recognition (exp 9).py - E:\Open Cv (11239A003)\Exp - 25\Text detection an...
File Edit Format Run Options Window Help
import cv2
import pytesseract
import os
pytesseract.pytesseract.tesseract_cmd = r"C:\Program Files\Tesseract-OCR\tesseract
img_path = r"E:\Open Cv (11239A003)\images.png"
if not os.path.exists(img_path):
print("Error: Image path is incorrect")
exit()
img = cv2.imread(img_path)
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
gray = cv2.medianBlur(gray, 3)
thresh = cv2.adaptiveThreshold(
gray, 255,
cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
cv2.THRESH_BINARY,
11, 2
)
custom_config = r'--oem 3 --psm 6'
text = pytesseract.image_to_string(thresh, config=custom_config)
print("Detected Text:")
print(text)
cv2.imshow("Original Image", img)
cv2.imshow("Processed Image", thresh)
cv2.waitKey(0)
cv2.destroyAllWindows()
Ln: 10 Col: 43
IDLE Shell 3.10.6
File Edit Shell Debug Options Window Help
Python 3.10.6 (tags/v3.10.6:9c7b4bd, Aug 1 2022, 21:53:49) [MSC v.1932 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: E:\Open Cv (11239A003)\Exp - 25\Text detection and recognition (exp 9).py
Detected Text:
a
(s\. SCSVMV
REY, Deemed to be University
SeeeE u/s of UGC act 1956
Original Image
SCSVMV
Deemed to be University
u/s of UGC act 1956
Processed Image
SCSVMV
Deemed to be University
u/s of UGC act 1956
Activate Windows
Go to Settings to activate Windows.
Ln: 5 Col: 0